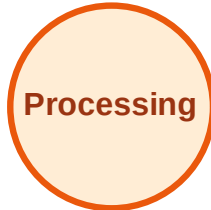


BFS Traversal

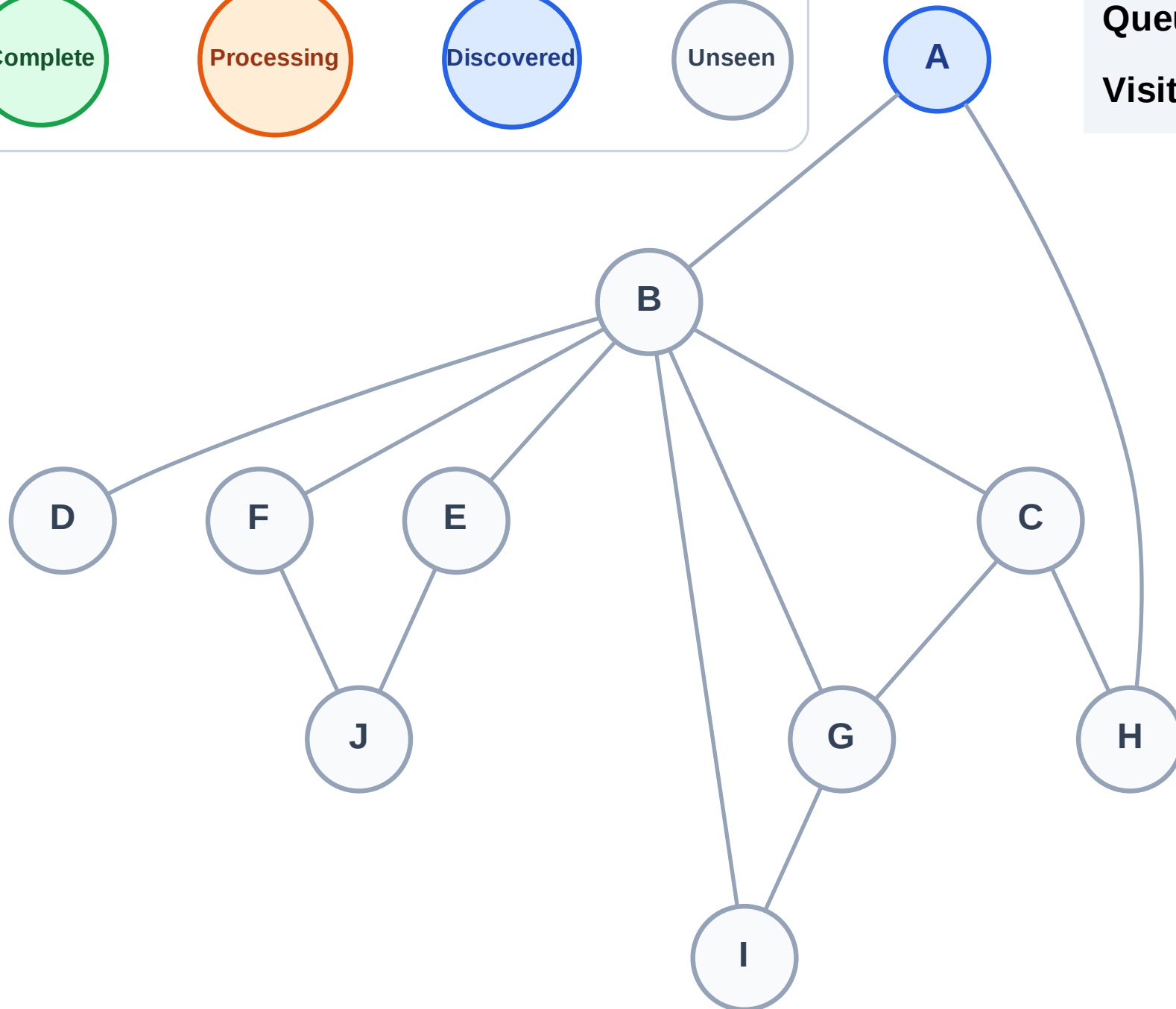
Step 1: Start at A

Legend



Queue: A

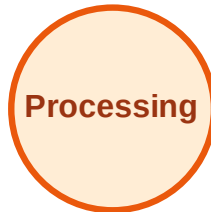
Visited: (none)



BFS Traversal

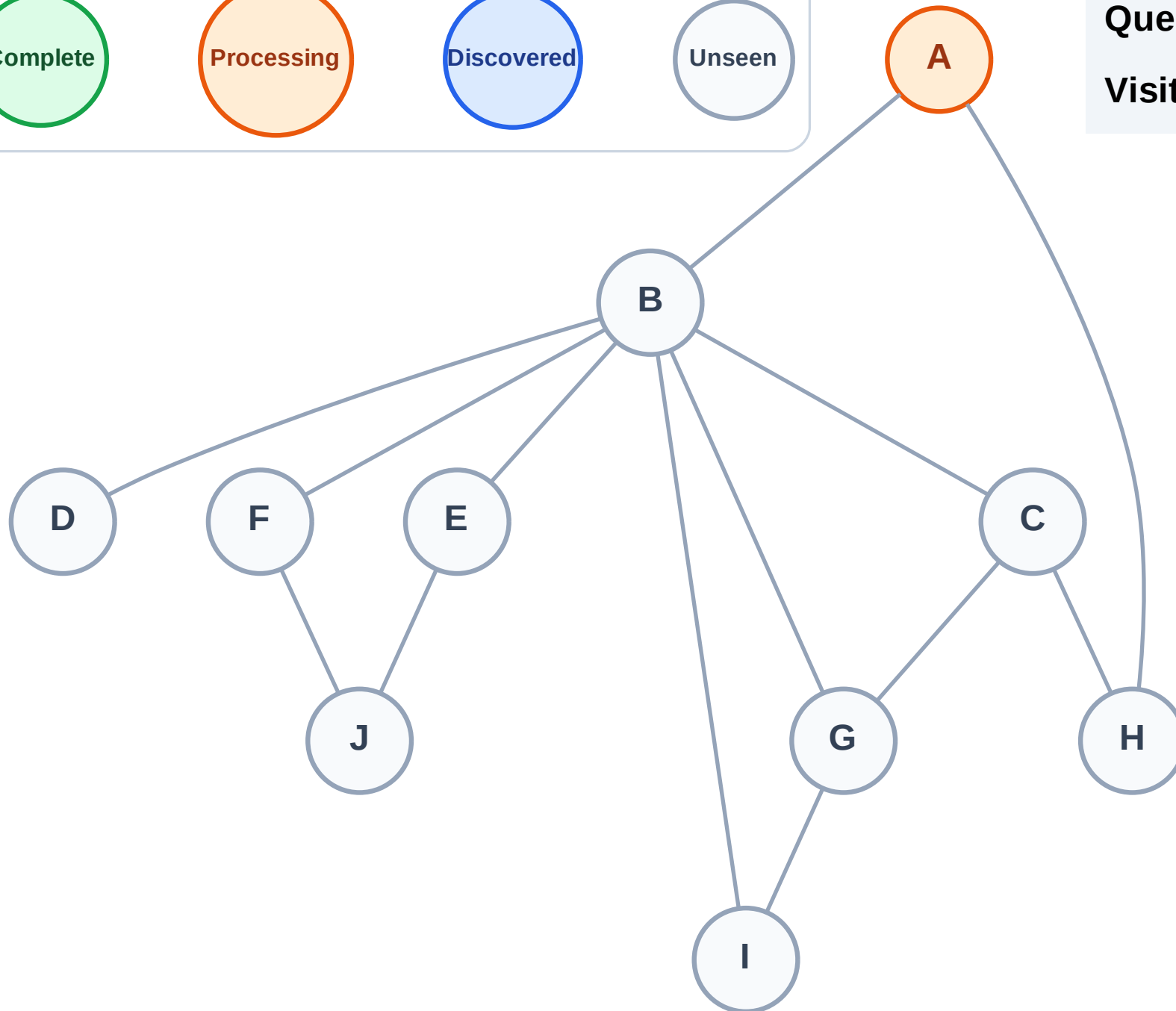
Step 2: Process node A

Legend



Queue: (empty)

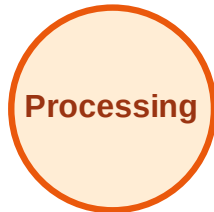
Visited: (none)



BFS Traversal

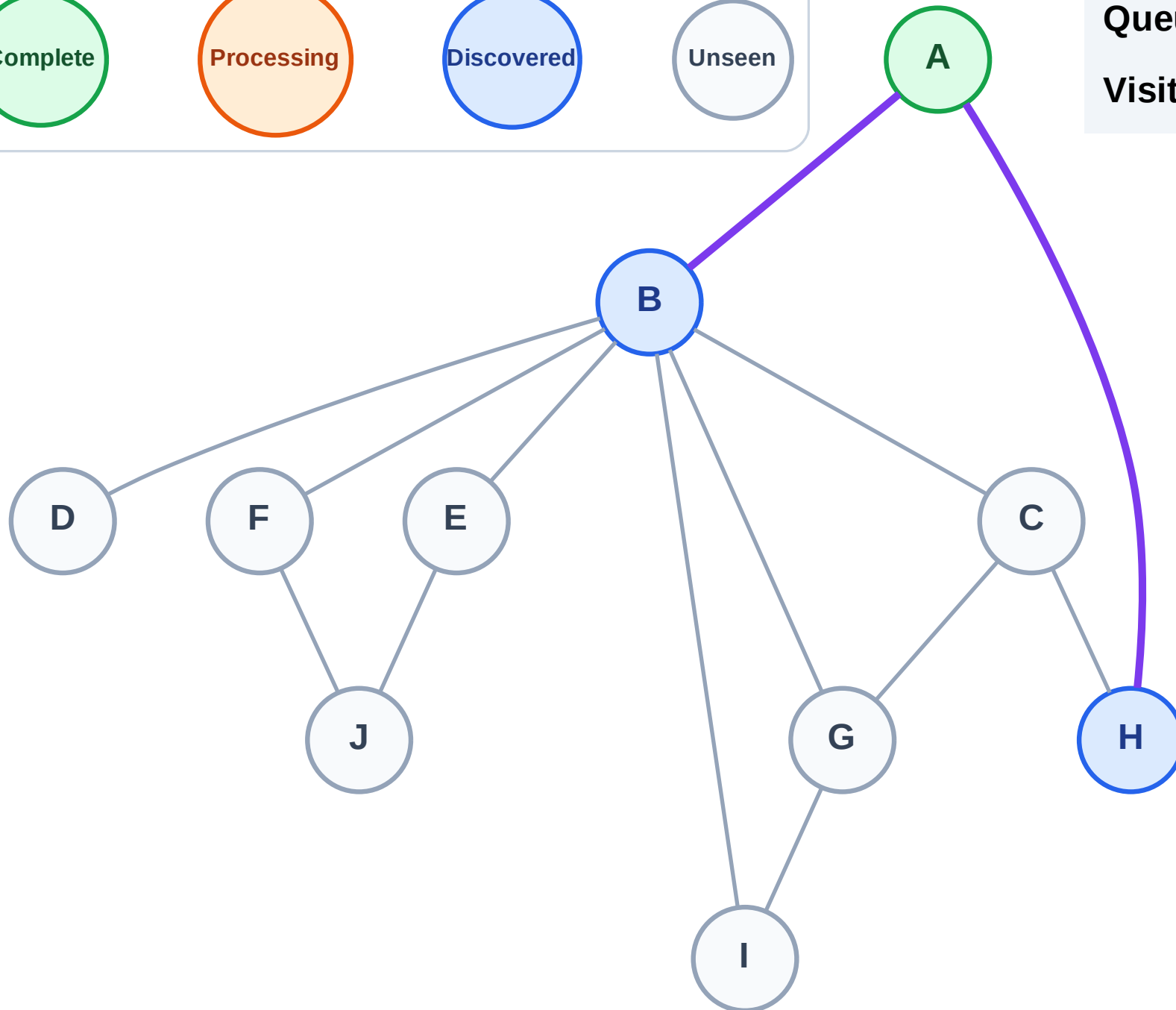
Step 3: Discover B, H from A

Legend



Queue: B → H

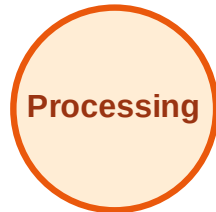
Visited: A



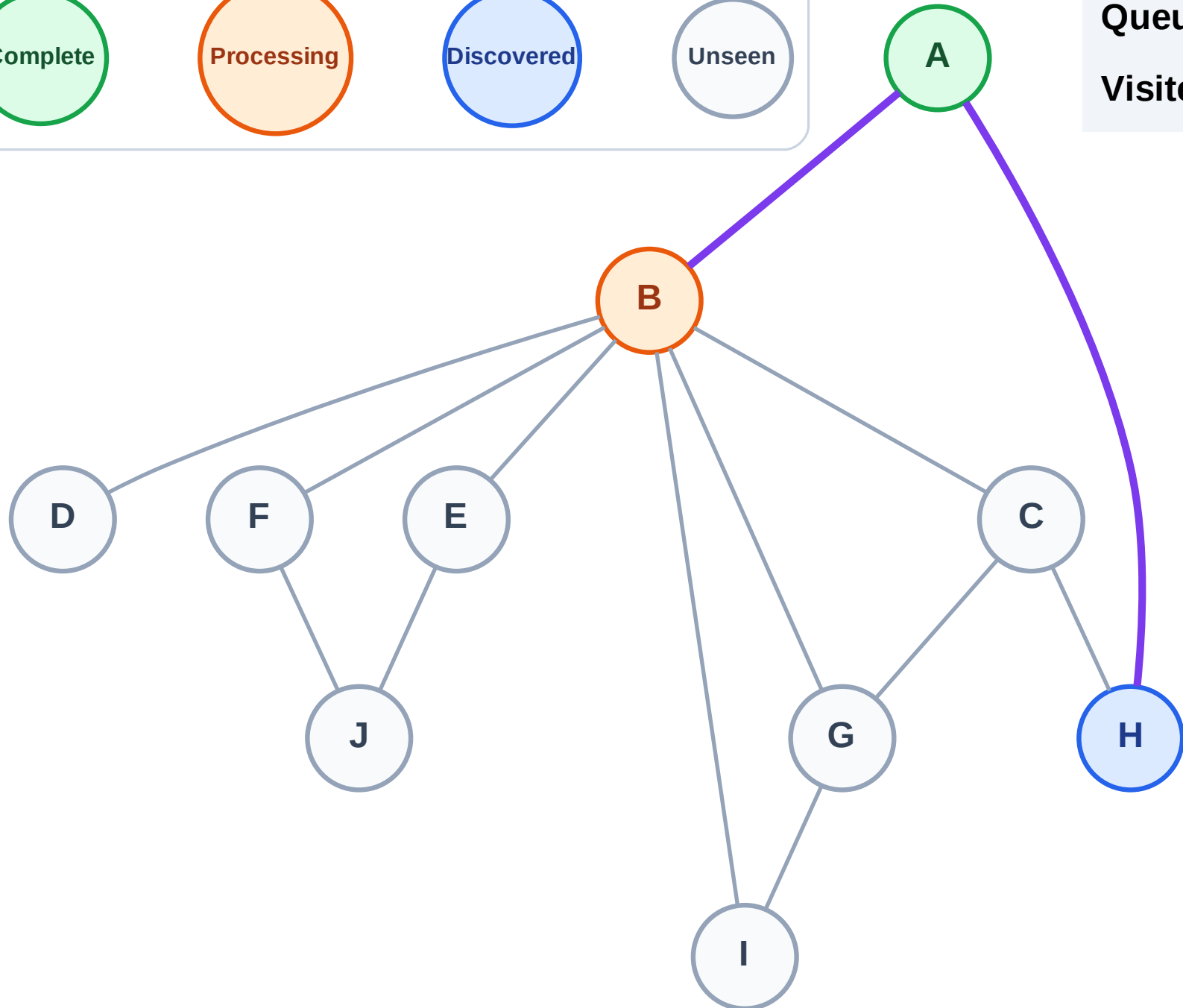
BFS Traversal

Step 4: Process node B

Legend



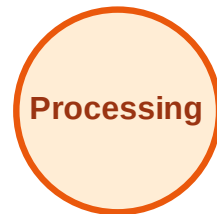
Queue: H
Visited: A



BFS Traversal

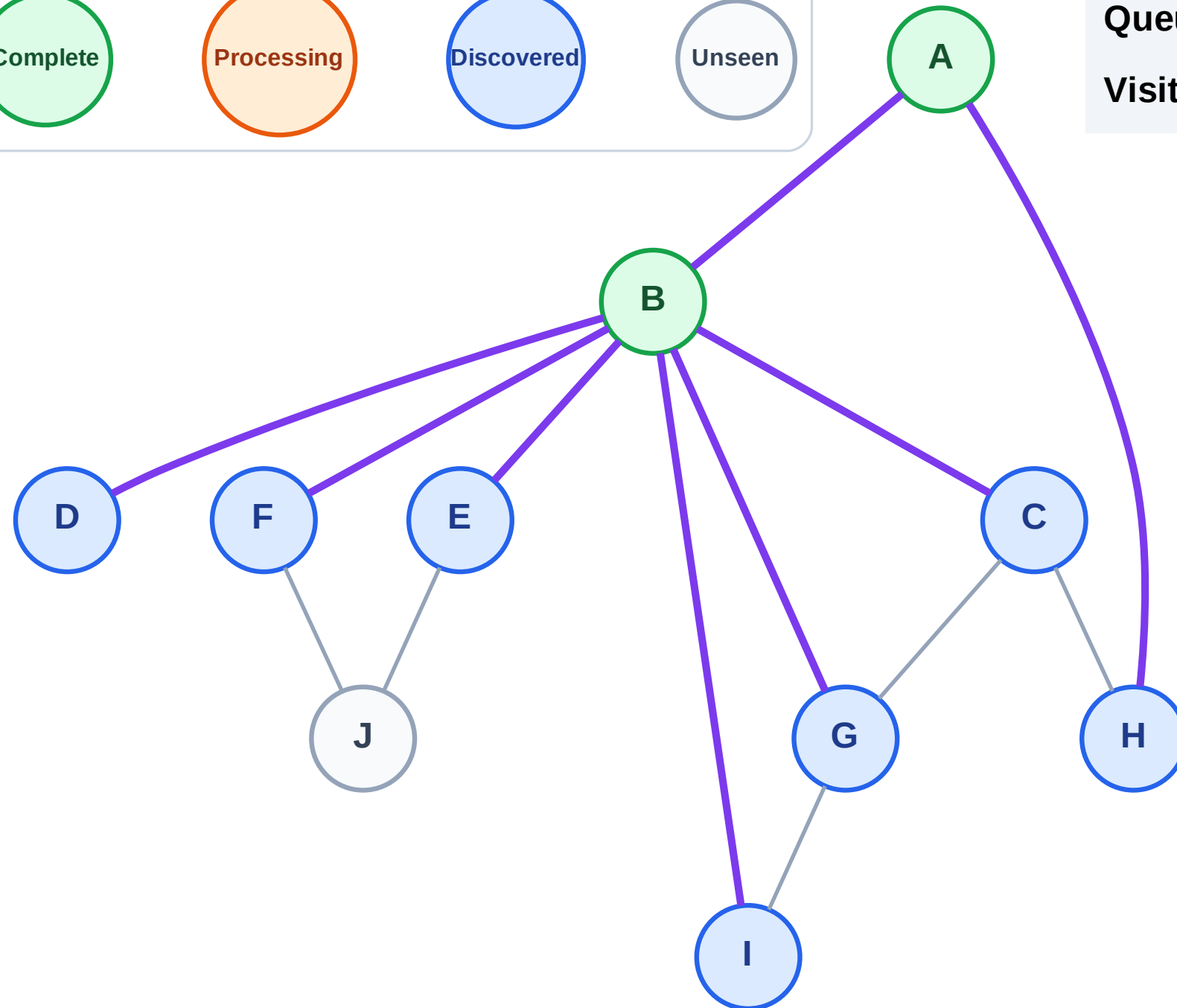
Step 5: Discover C, D, E, F, G, I from B

Legend



Queue: H → C → D → E → F → G → I

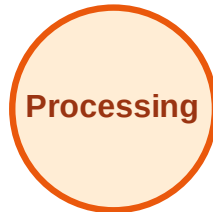
Visited: A, B



BFS Traversal

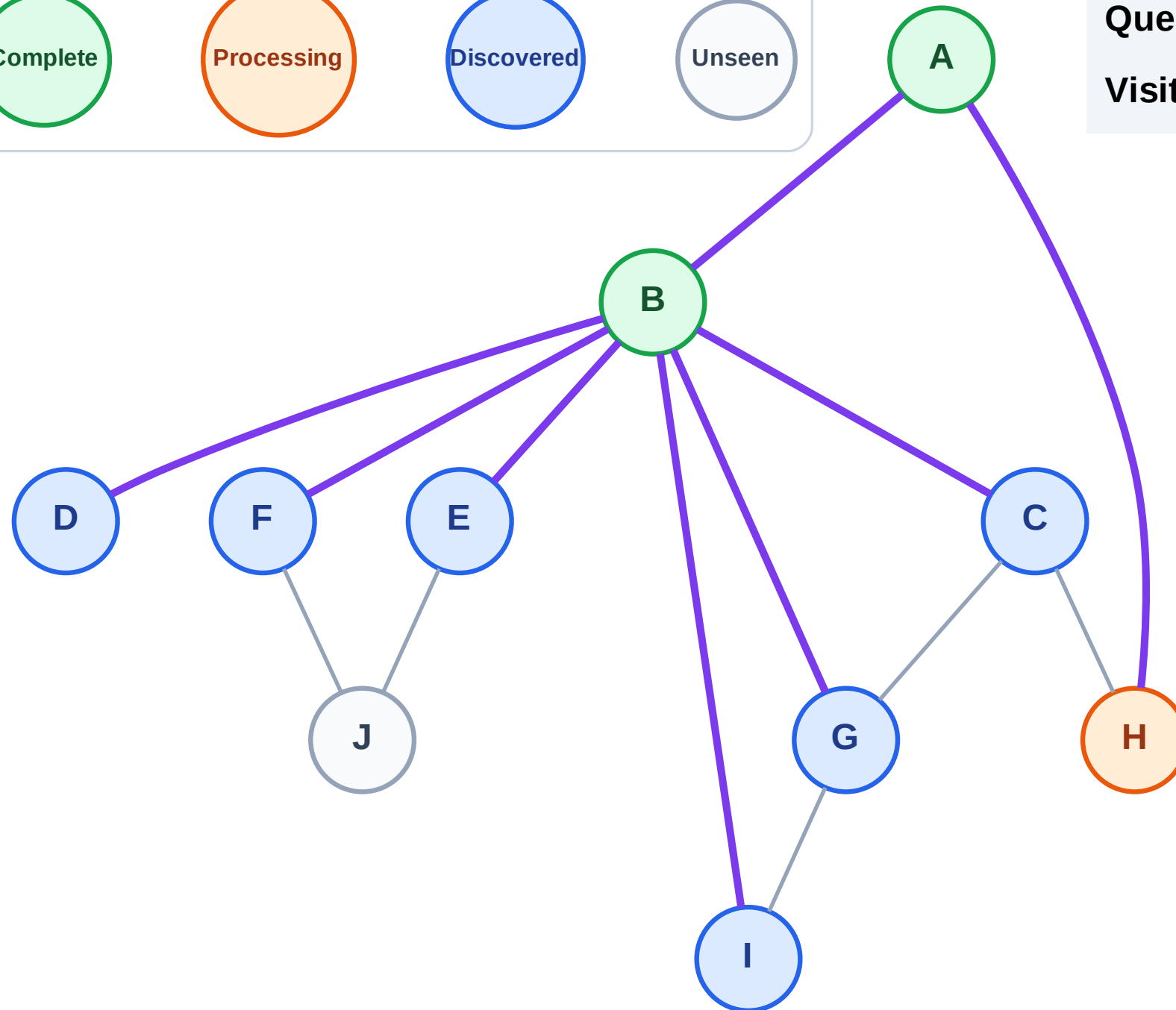
Step 6: Process node H

Legend



Queue: C → D → E → F → G → I

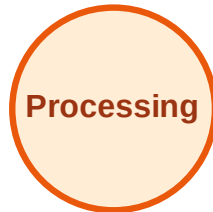
Visited: A, B



BFS Traversal

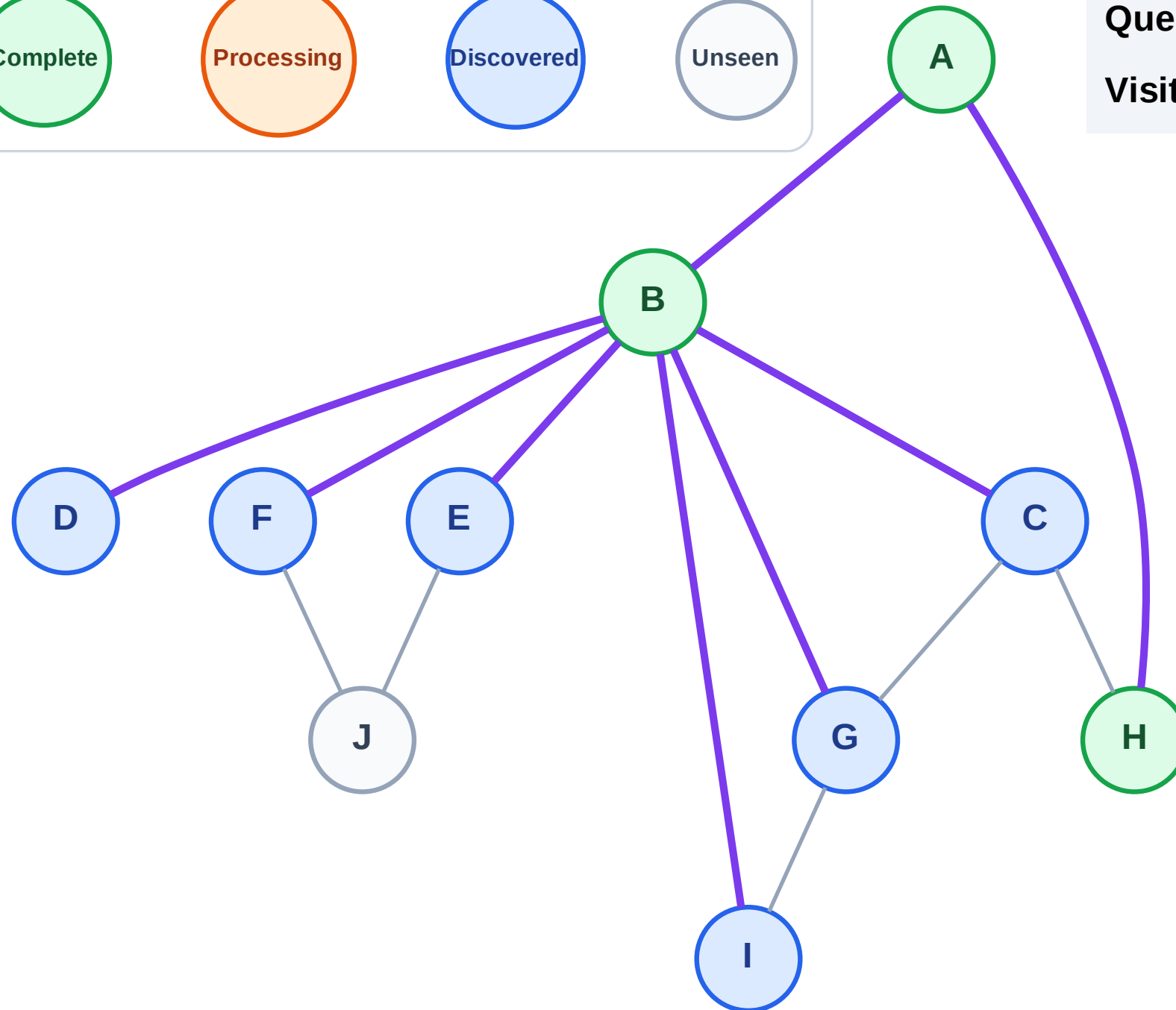
Step 7: No new neighbors from H

Legend



Queue: C → D → E → F → G → I

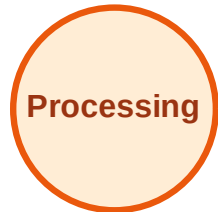
Visited: A, B, H



BFS Traversal

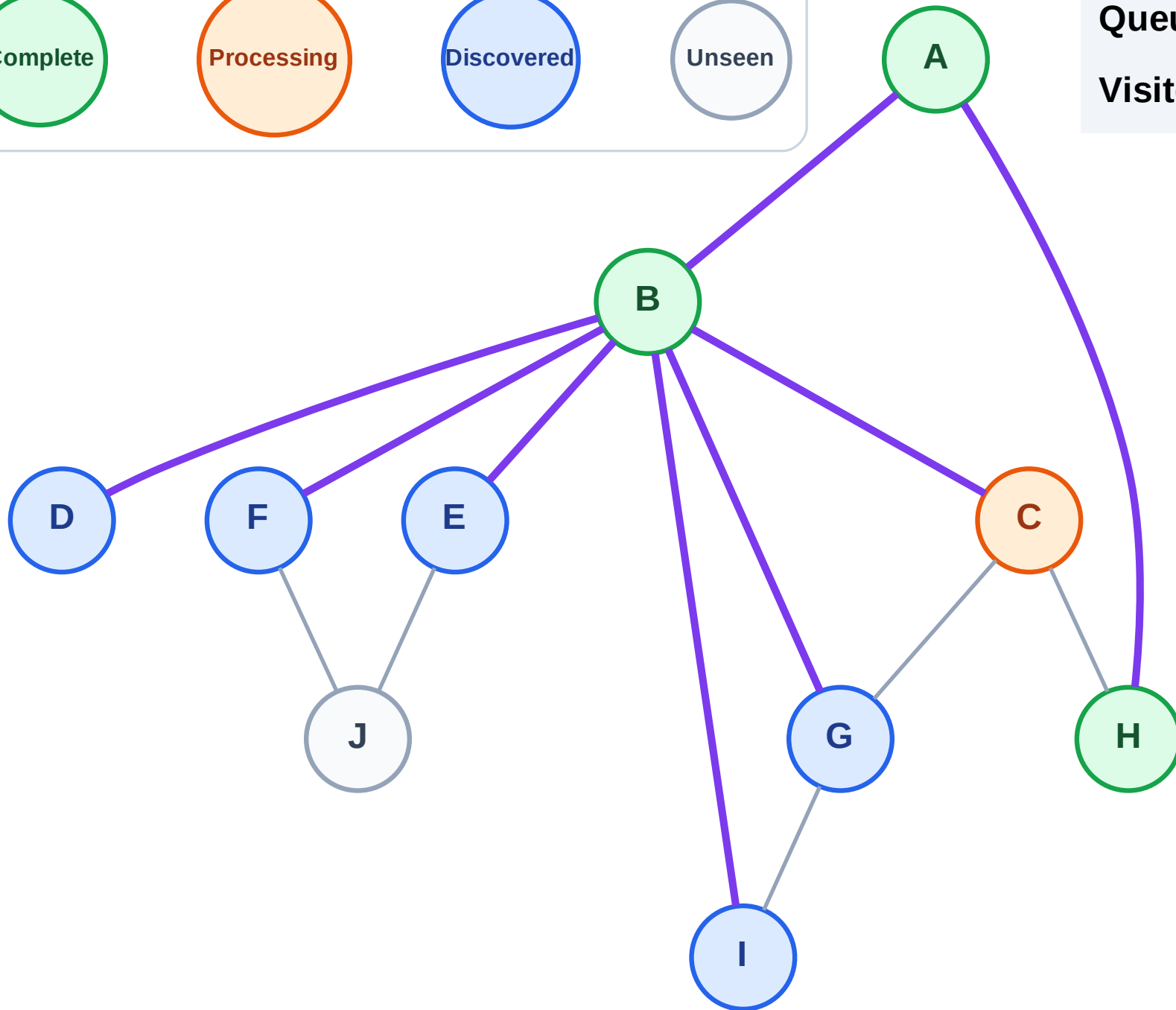
Step 8: Process node C

Legend



Queue: D → E → F → G → I

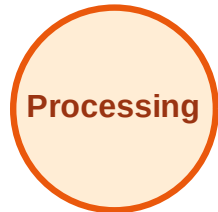
Visited: A, B, H



BFS Traversal

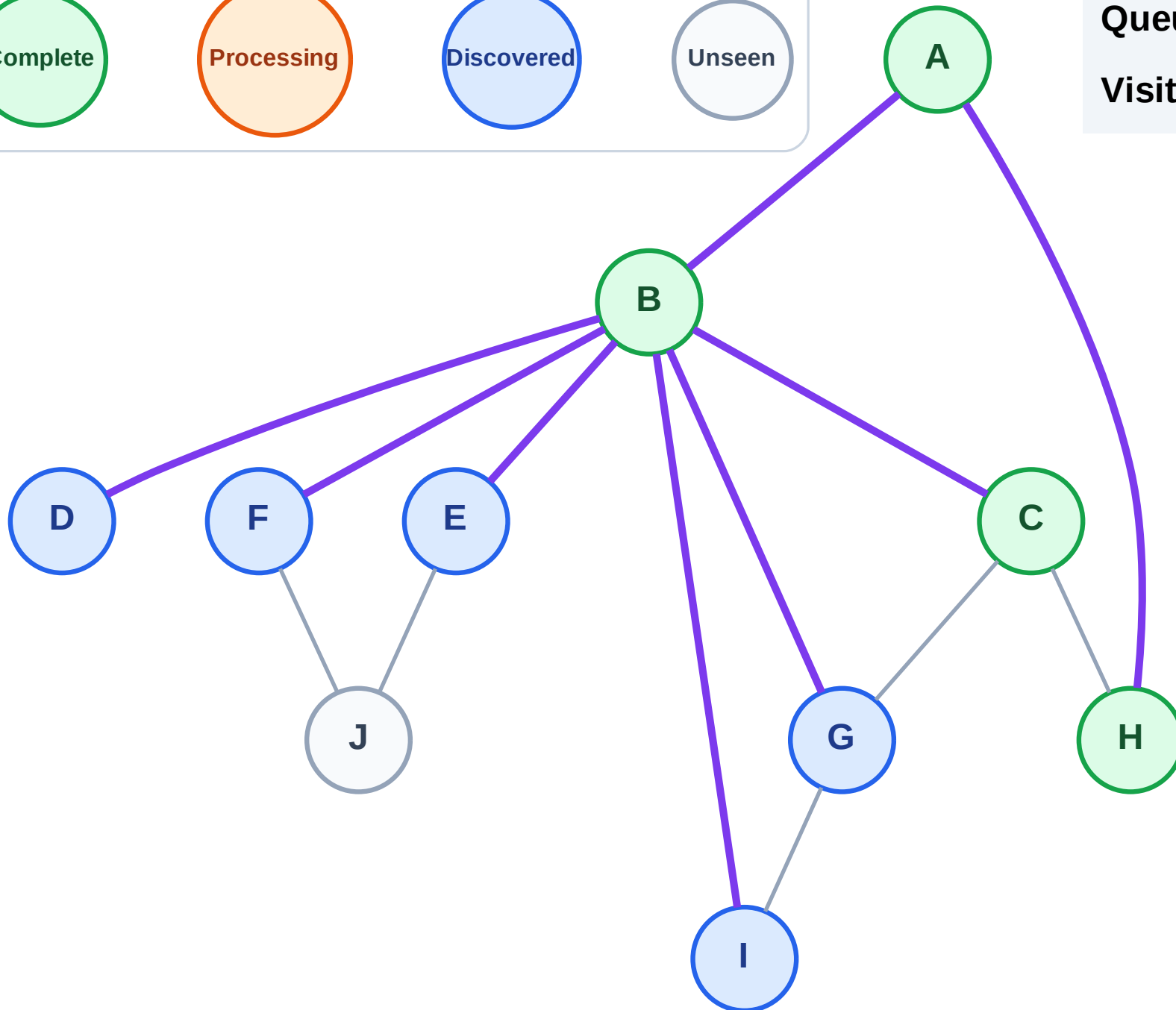
Step 9: No new neighbors from C

Legend



Queue: D → E → F → G → I

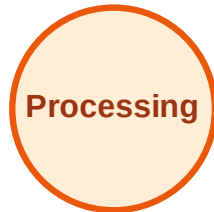
Visited: A, B, C, H



BFS Traversal

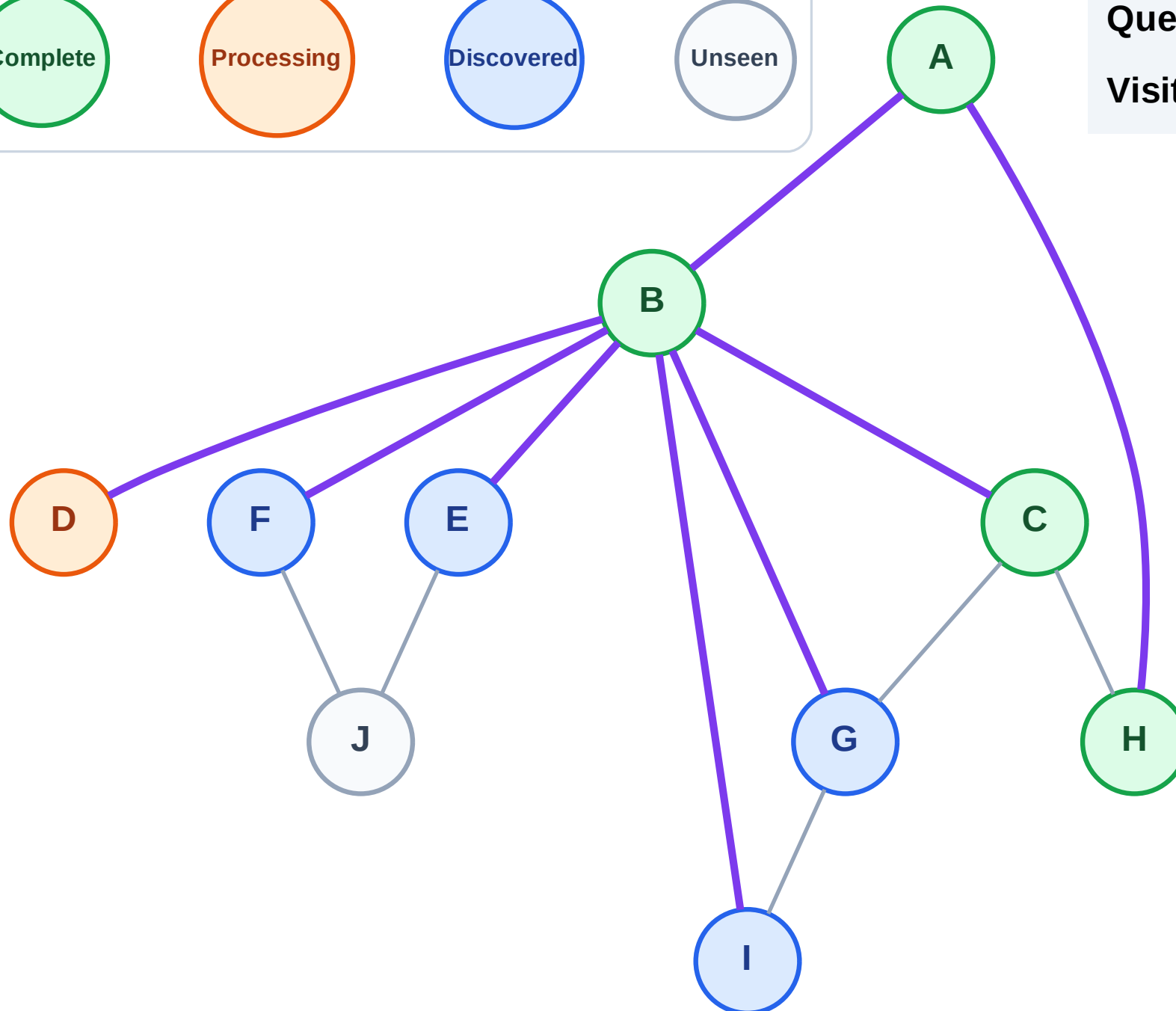
Step 10: Process node D

Legend



Queue: E → F → G → I

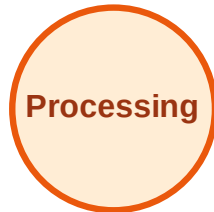
Visited: A, B, C, H



BFS Traversal

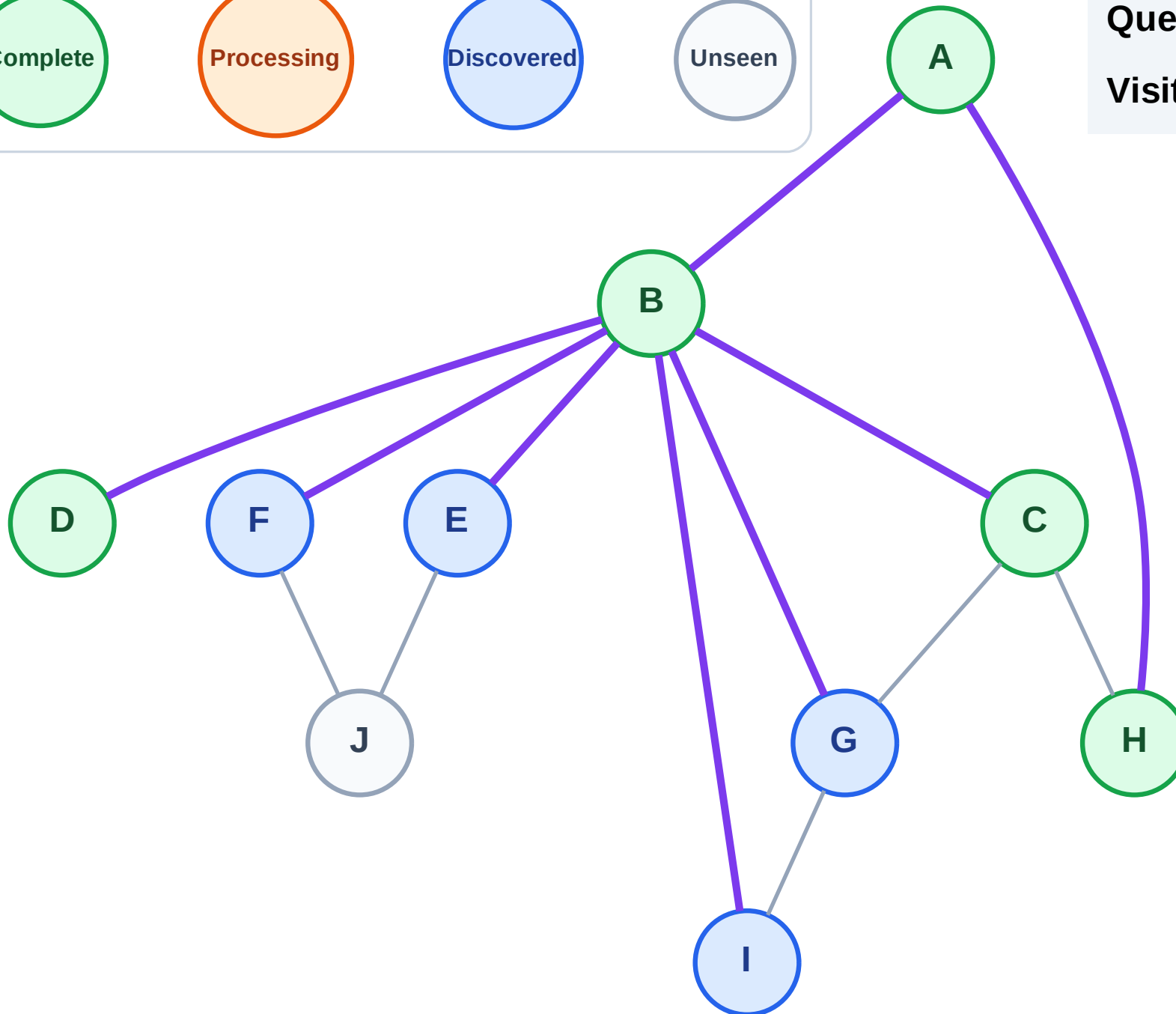
Step 11: No new neighbors from D

Legend



Queue: E → F → G → I

Visited: A, B, C, D, H



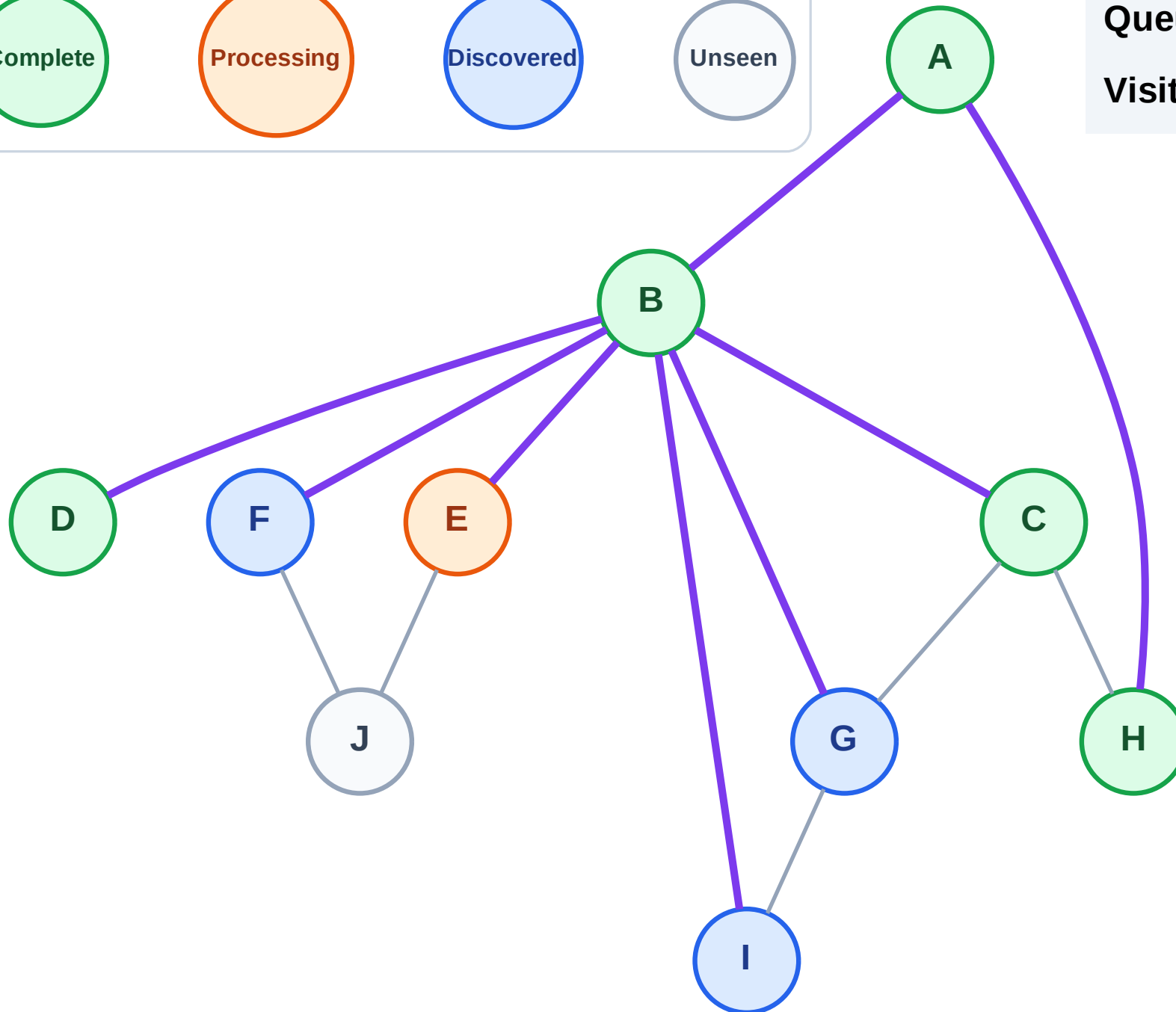
BFS Traversal

Step 12: Process node E

Legend



Queue: F → G → I
Visited: A, B, C, D, H



BFS Traversal

Step 13: Discover J from E

Legend

Complete

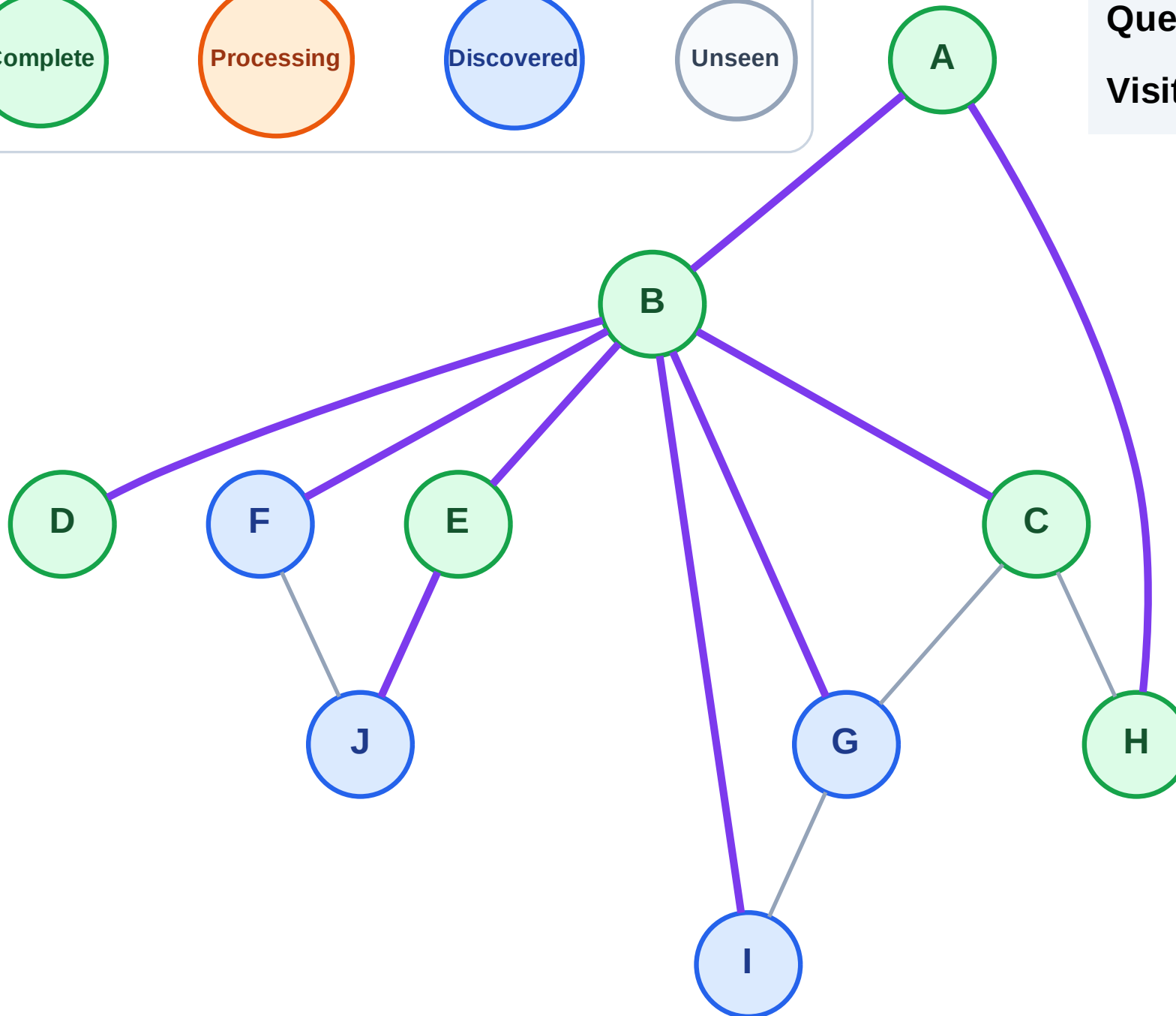
Processing

Discovered

Unseen

Queue: F → G → I → J

Visited: A, B, C, D, E, H



BFS Traversal

Step 14: Process node F

Legend

Complete

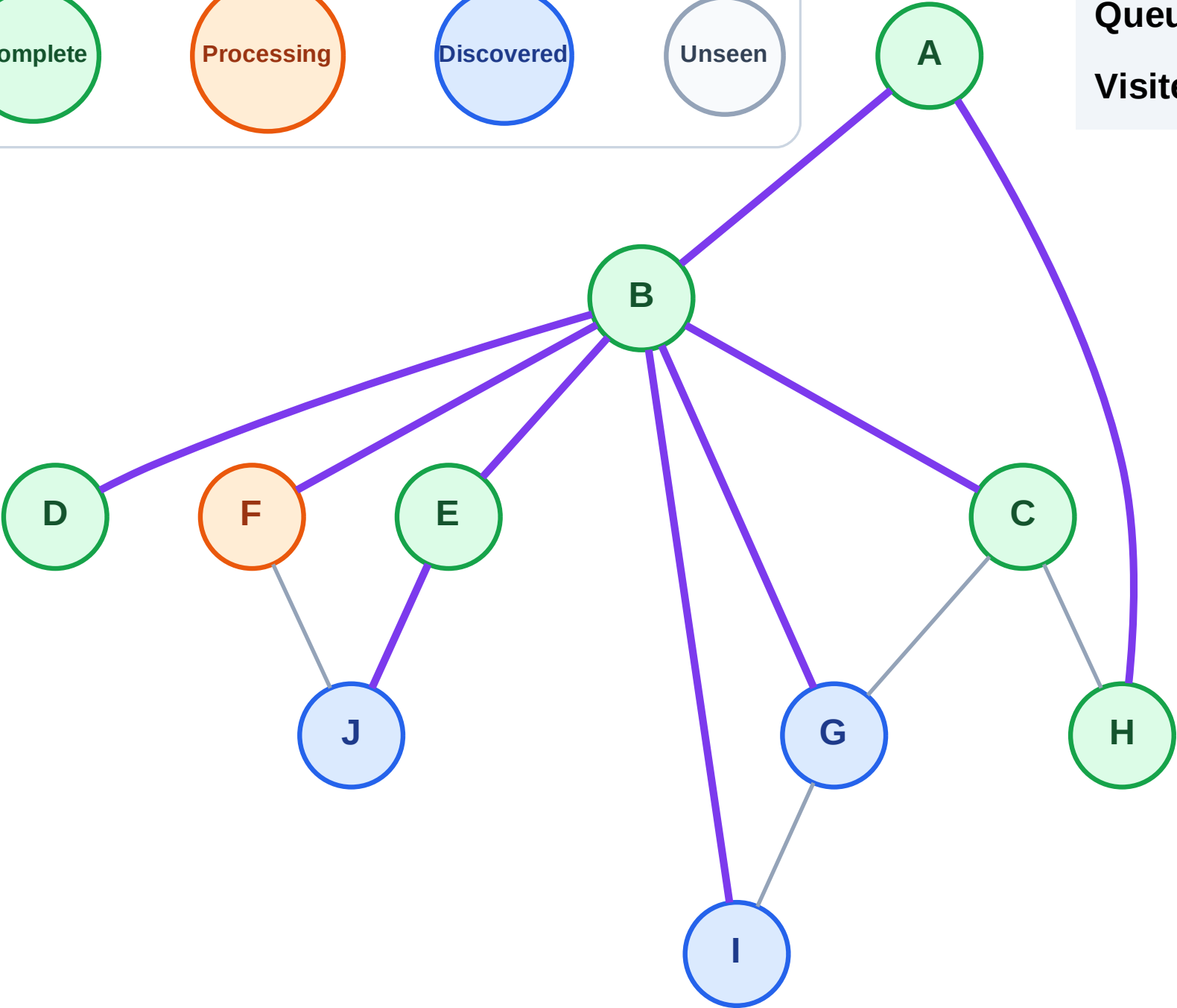
Processing

Discovered

Unseen

Queue: G → I → J

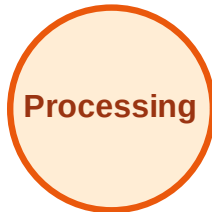
Visited: A, B, C, D, E, H



BFS Traversal

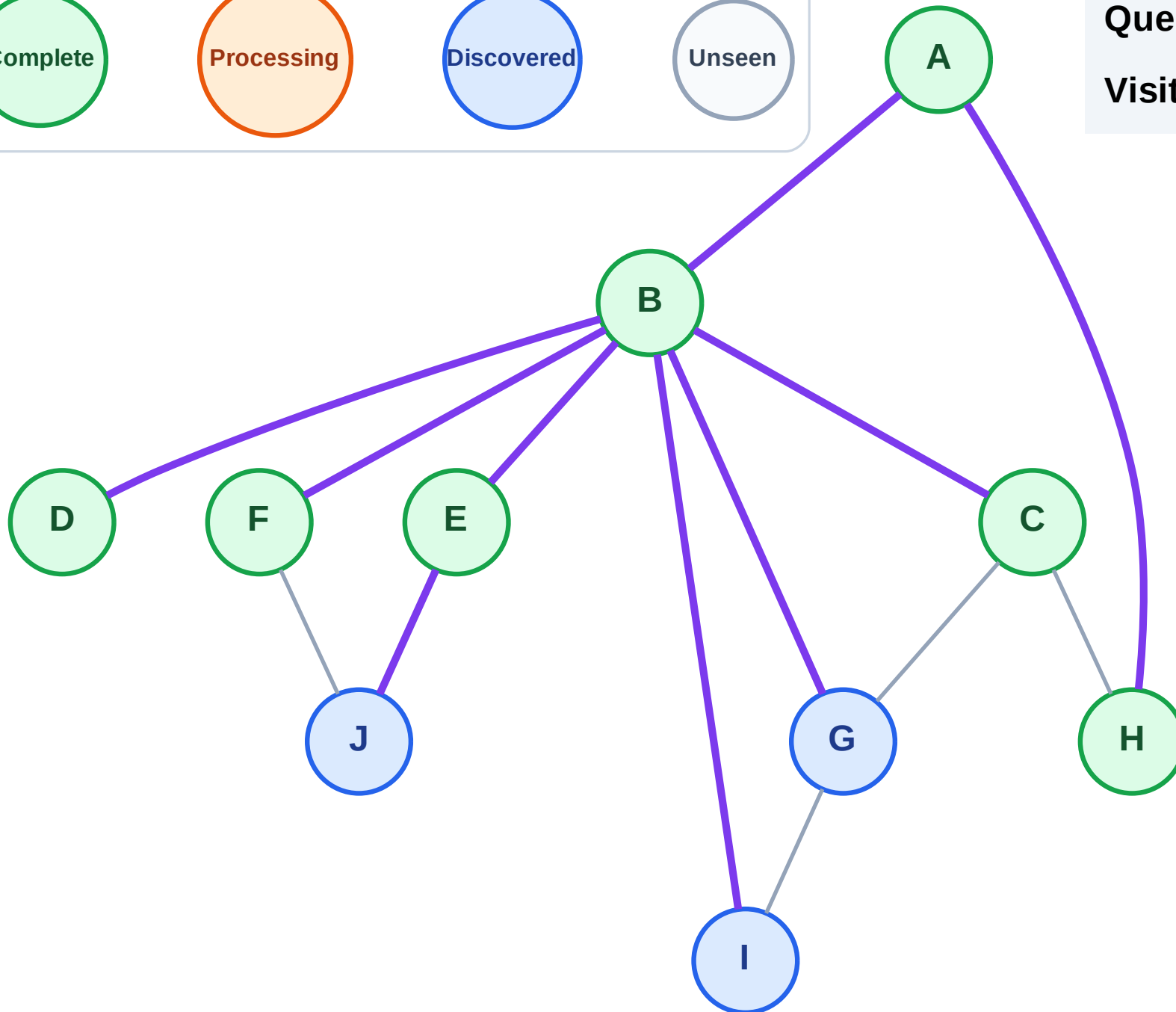
Step 15: No new neighbors from F

Legend



Queue: G → I → J

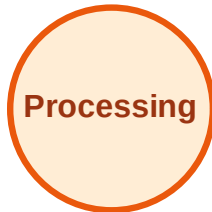
Visited: A, B, C, D, E, F, H



BFS Traversal

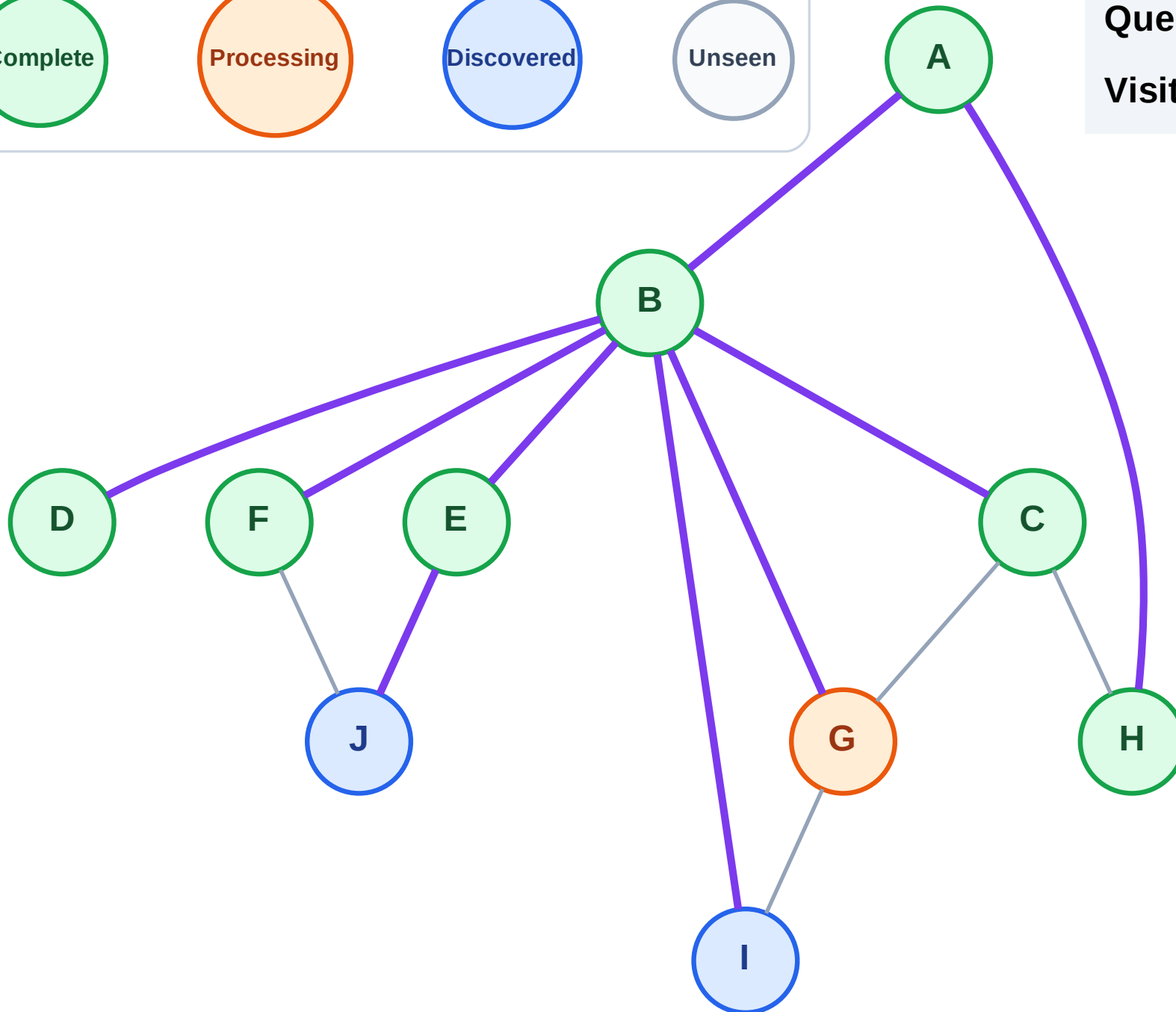
Step 16: Process node G

Legend



Queue: I → J

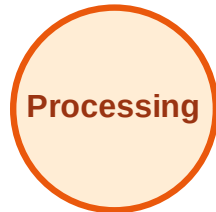
Visited: A, B, C, D, E, F, H



BFS Traversal

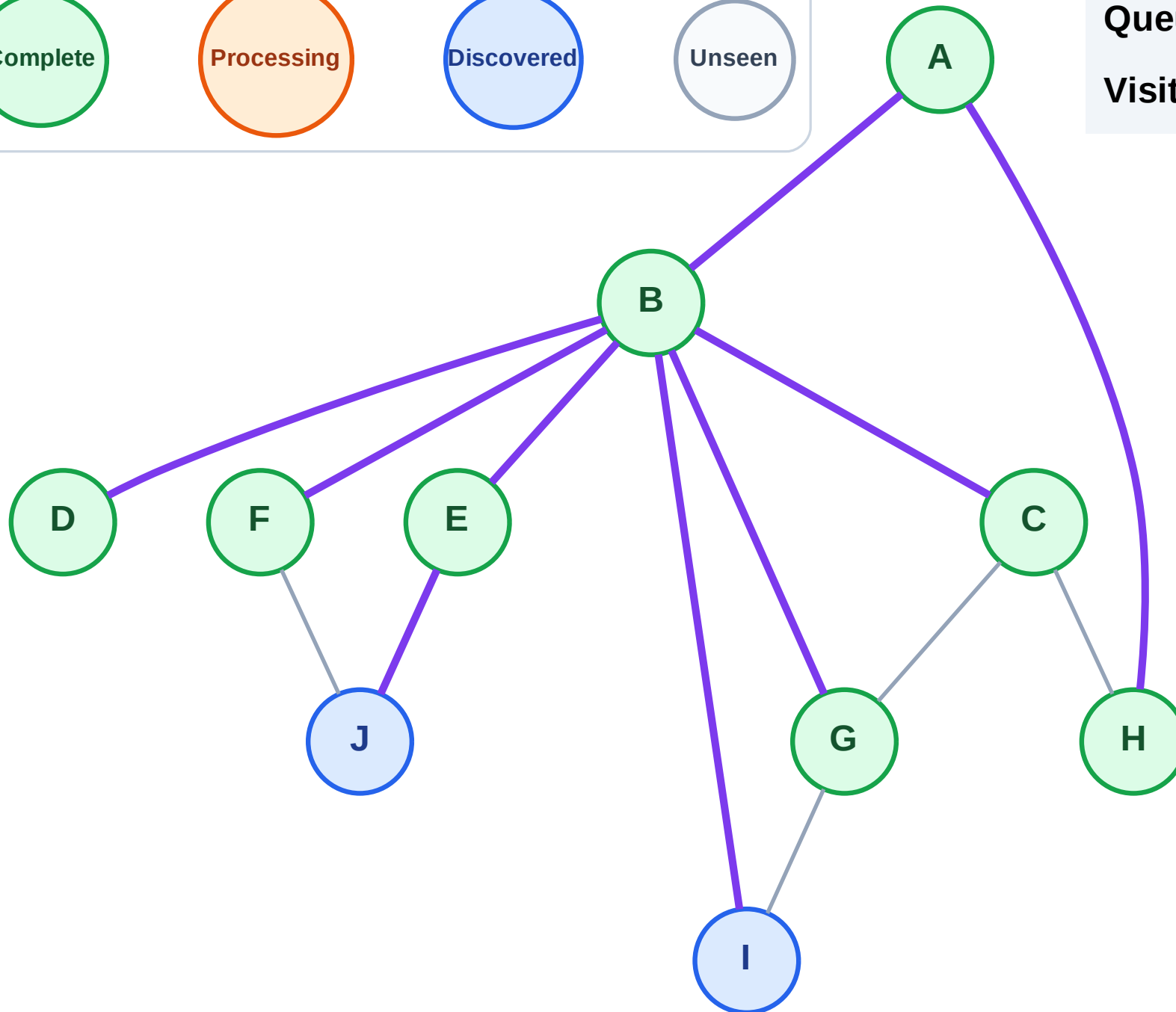
Step 17: No new neighbors from G

Legend



Queue: I → J

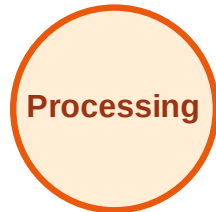
Visited: A, B, C, D, E, F, G, H



BFS Traversal

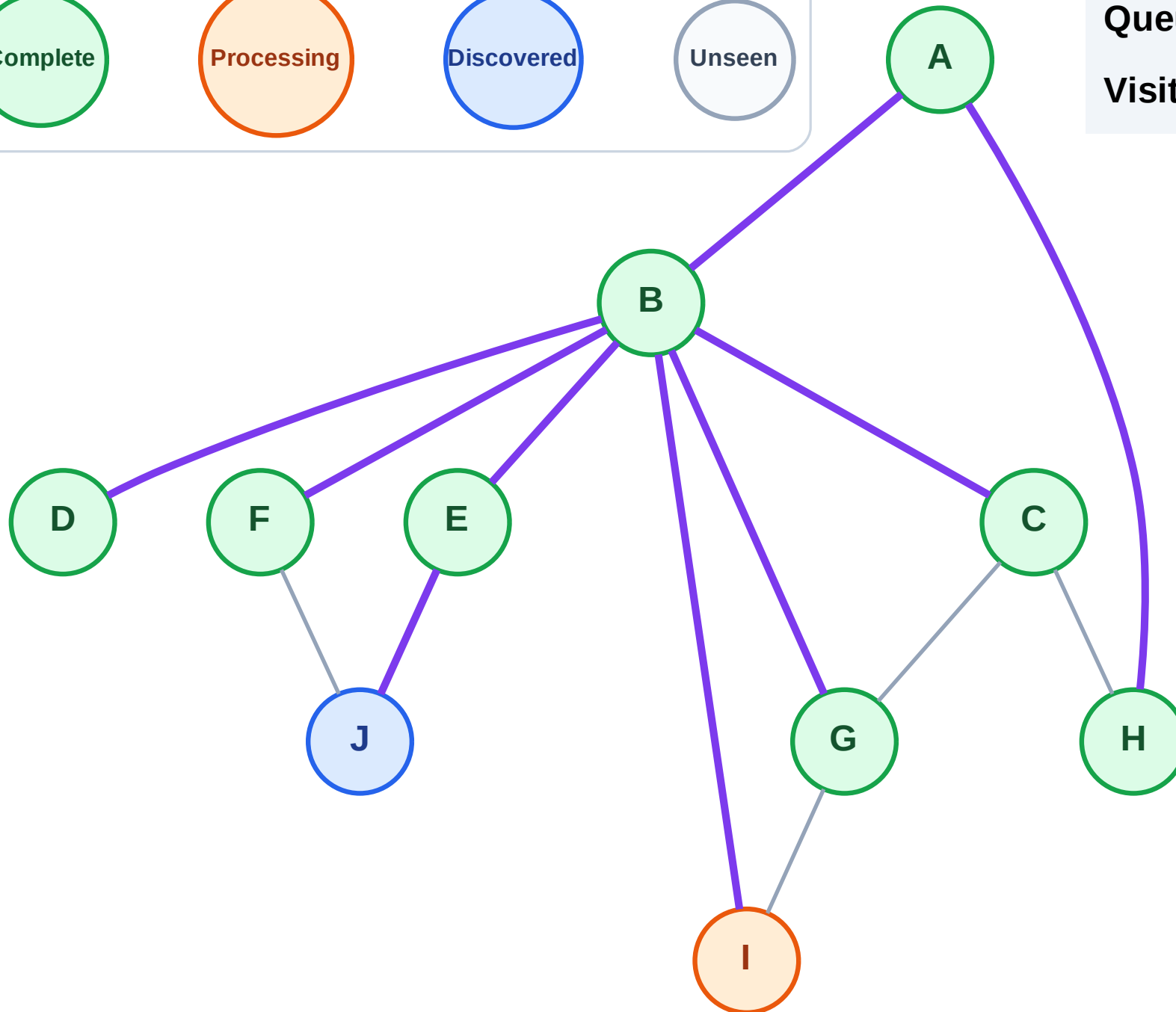
Step 18: Process node I

Legend



Queue: J

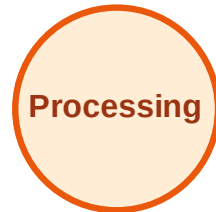
Visited: A, B, C, D, E, F, G, H



BFS Traversal

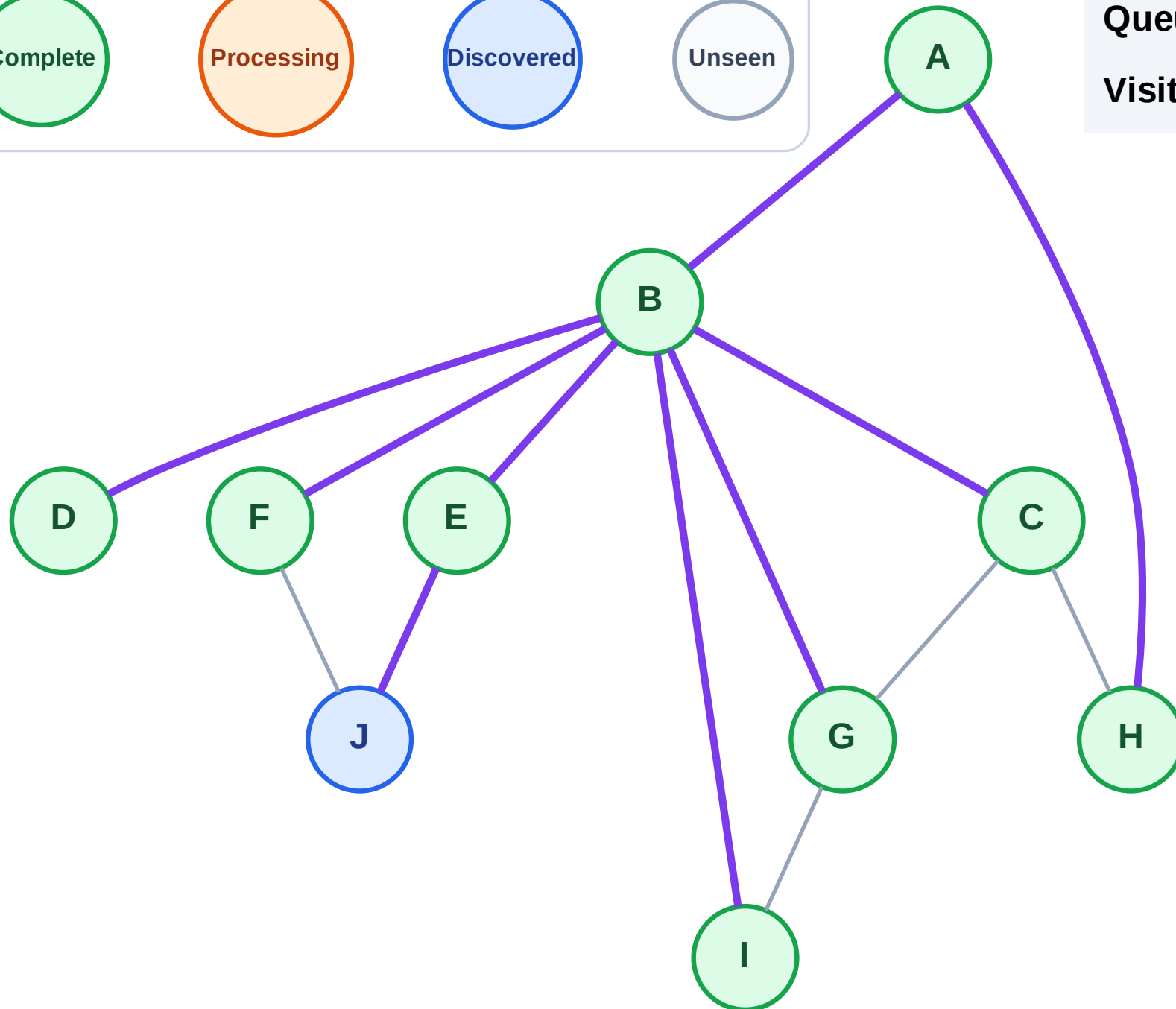
Step 19: No new neighbors from I

Legend



Queue: J

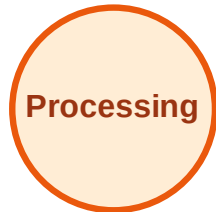
Visited: A, B, C, D, E, F, G, H, I



BFS Traversal

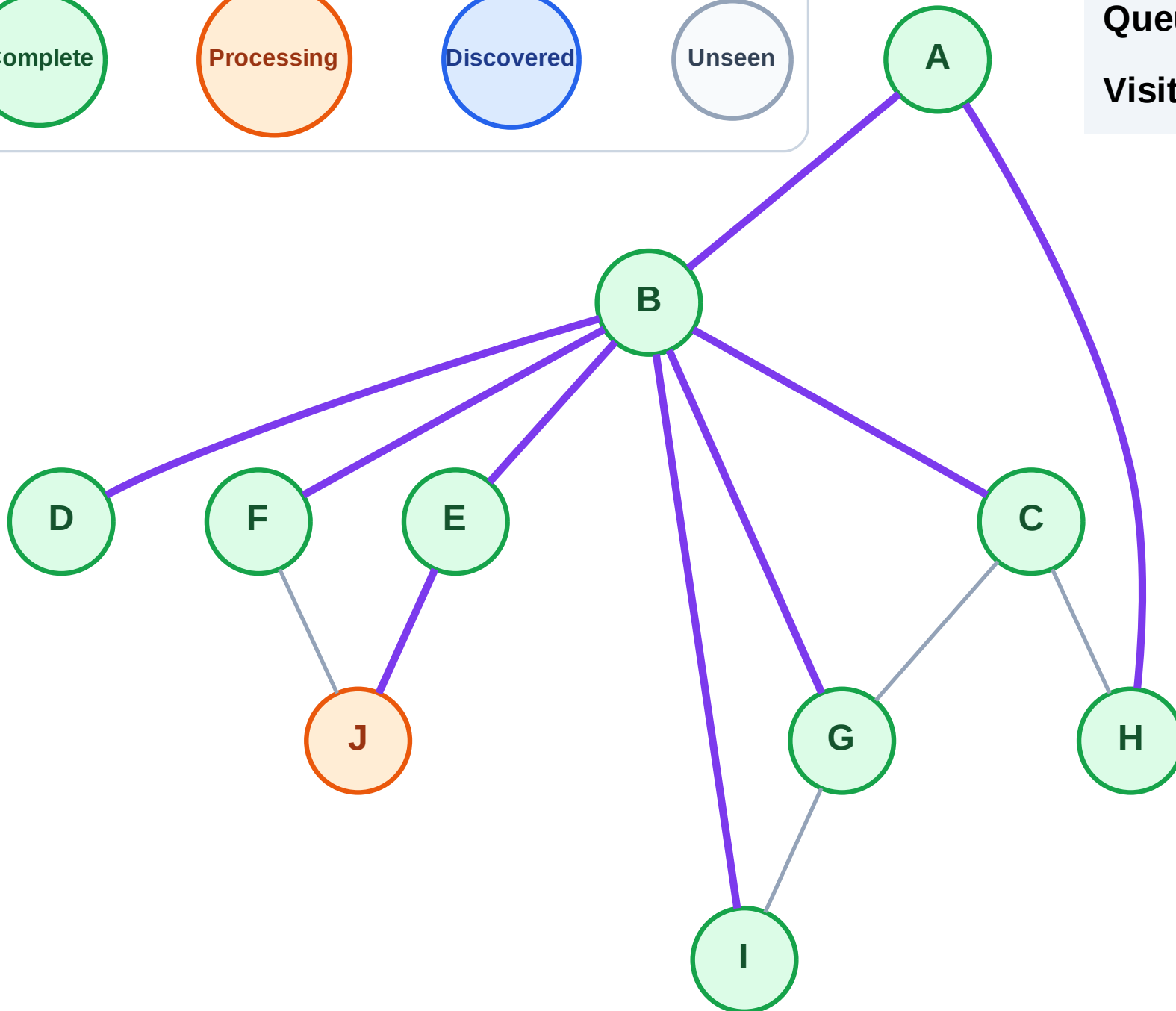
Step 20: Process node J

Legend



Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I



BFS Traversal

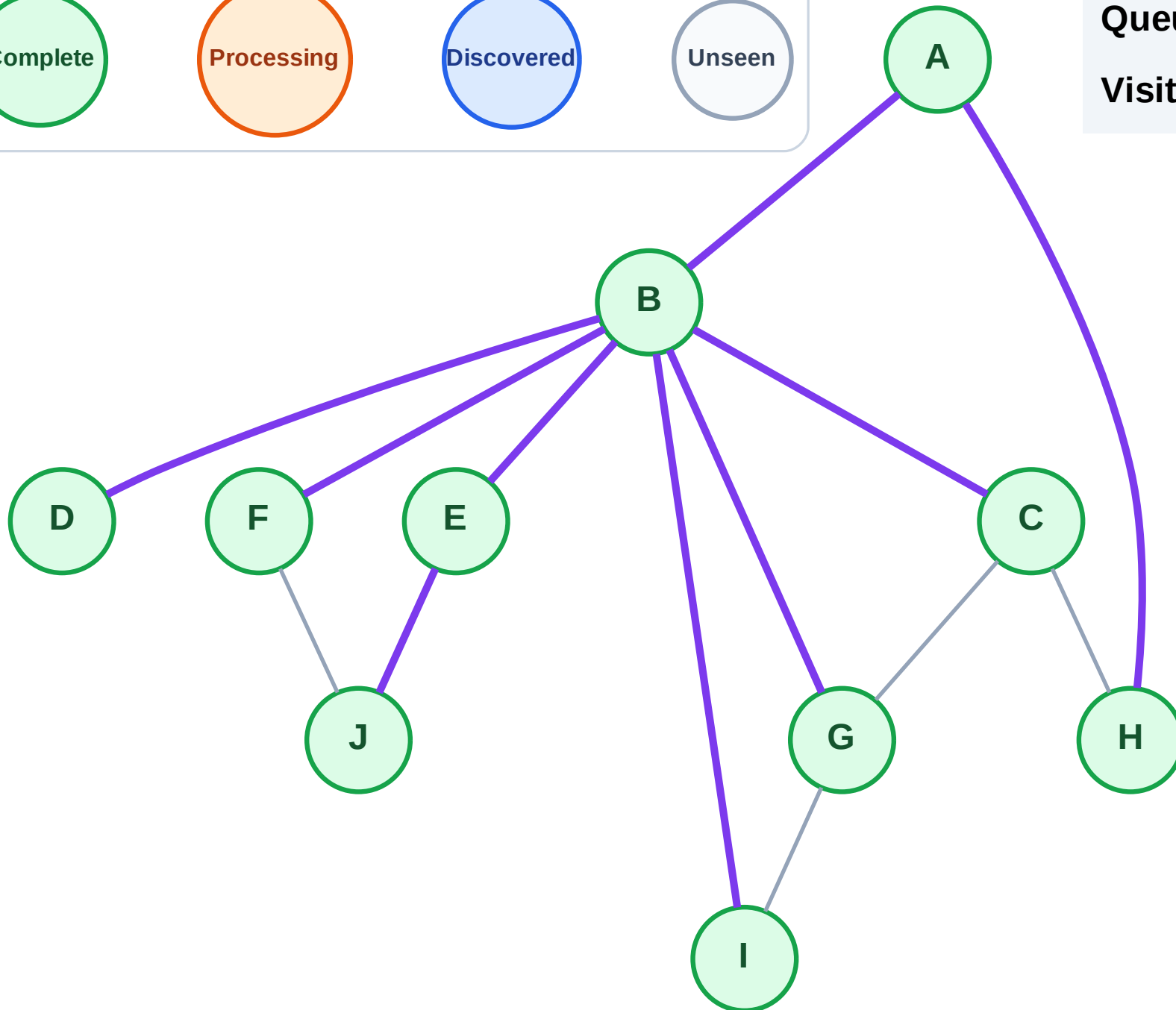
Step 21: No new neighbors from J

Legend



Queue: (empty)

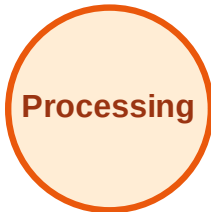
Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

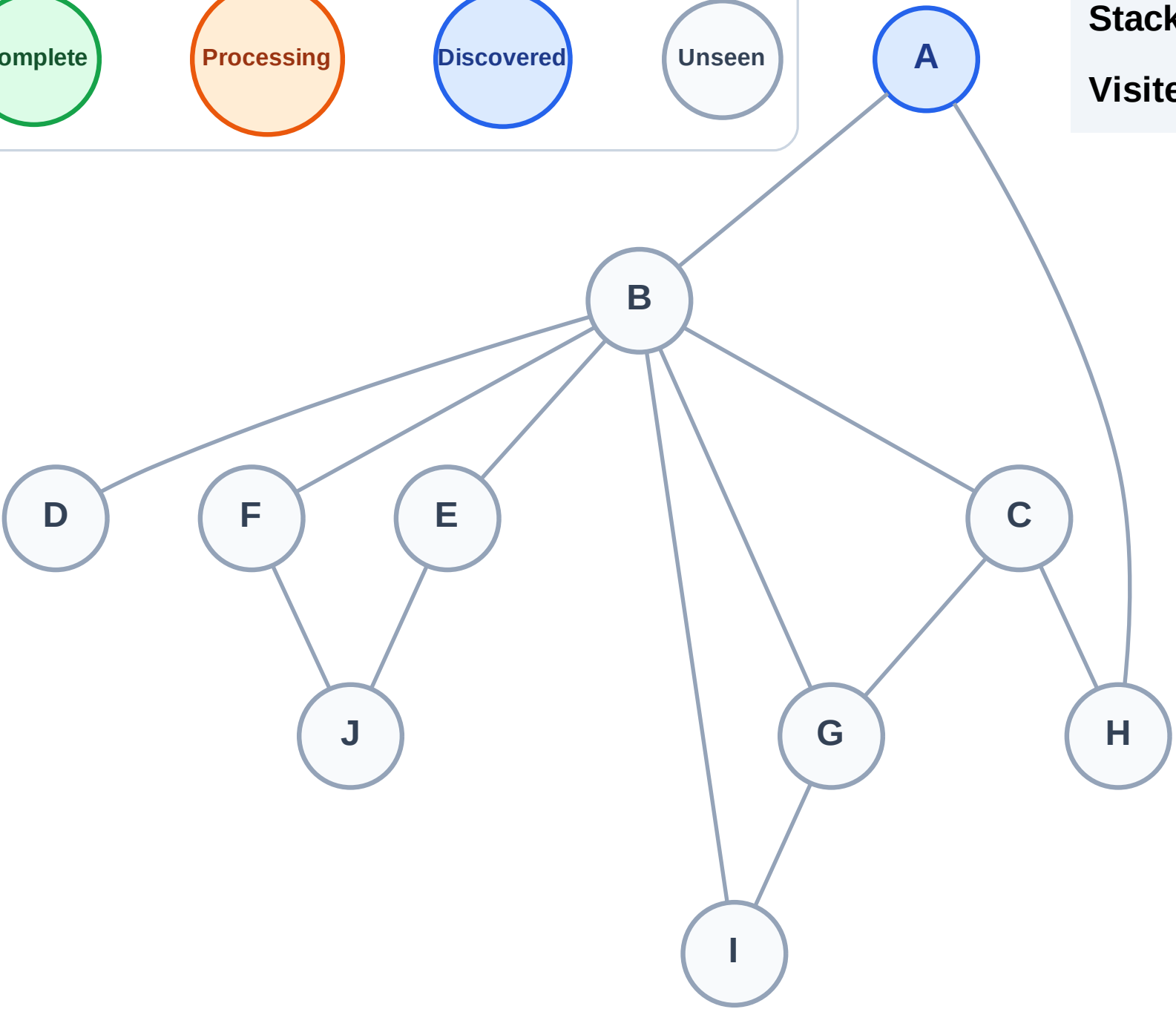
Step 1: Start at A

Legend



Stack (top at right): A

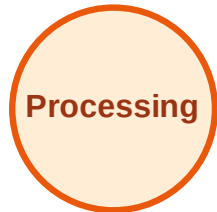
Visited: (none)



DFS Traversal

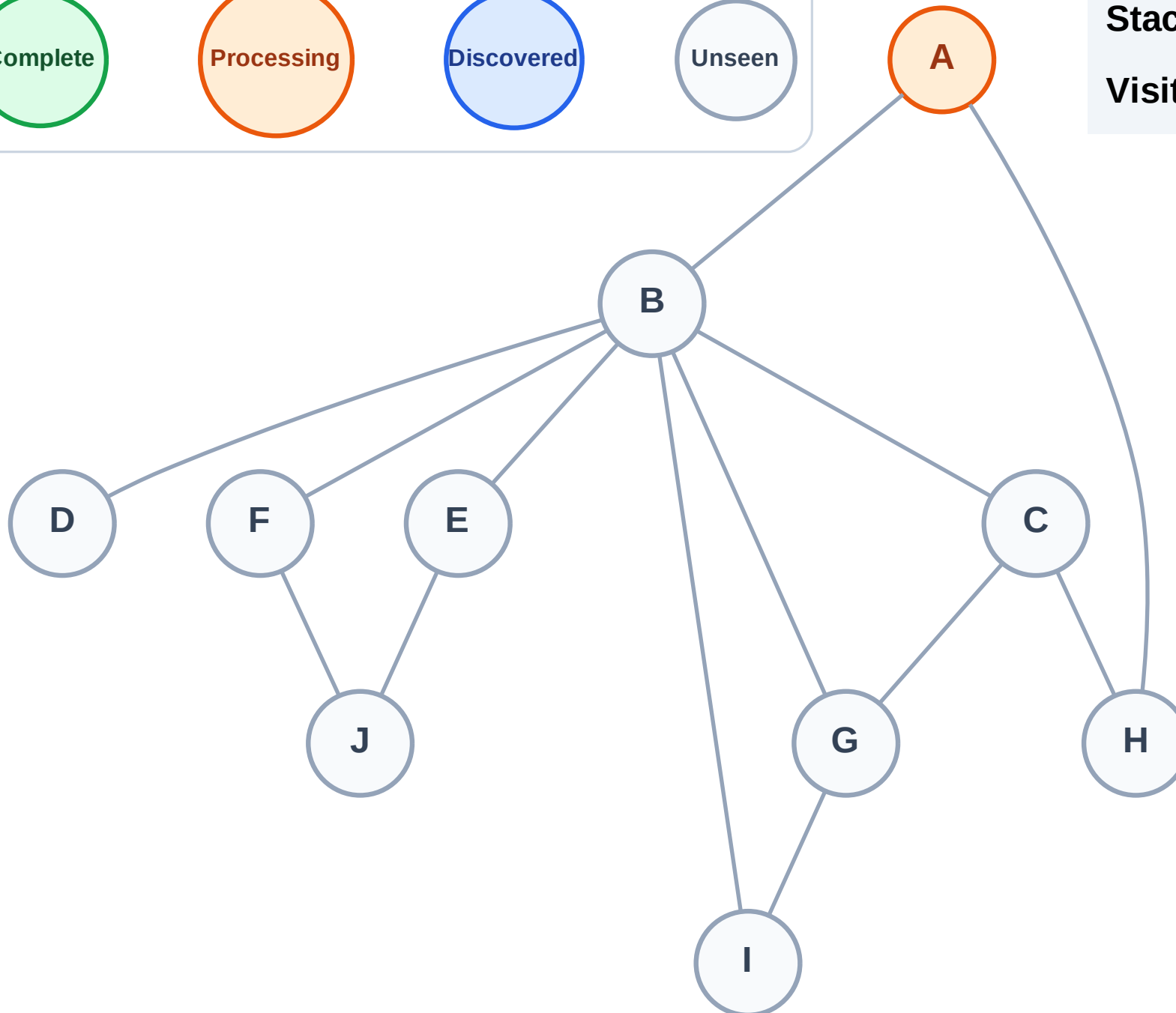
Step 2: Process node A

Legend



Stack (top at right): (empty)

Visited: (none)



DFS Traversal

Step 3: Discover H, B from A

Legend

Complete

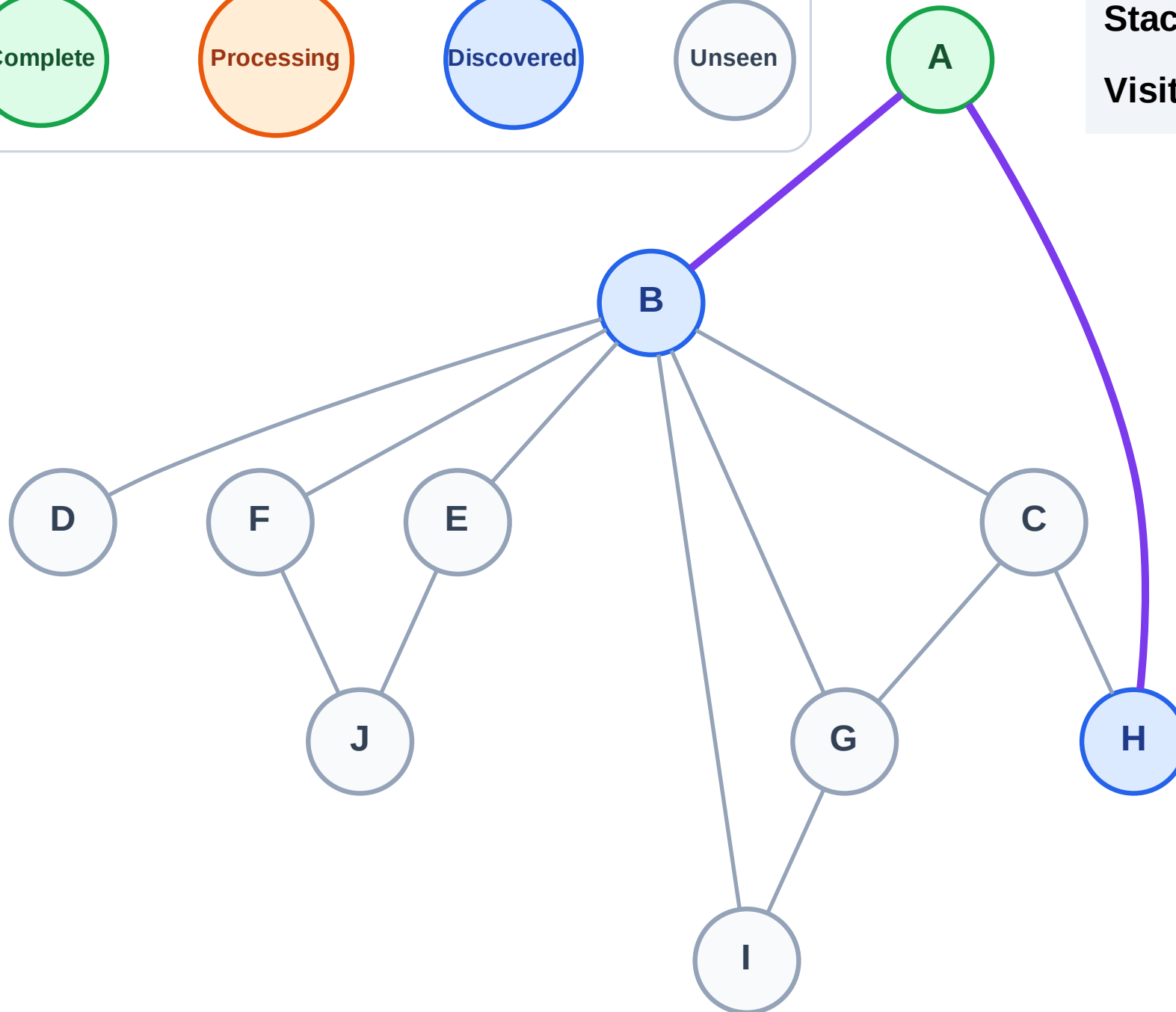
Processing

Discovered

Unseen

Stack (top at right): H | B

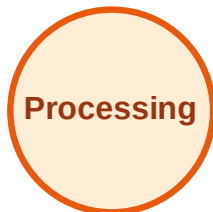
Visited: A



DFS Traversal

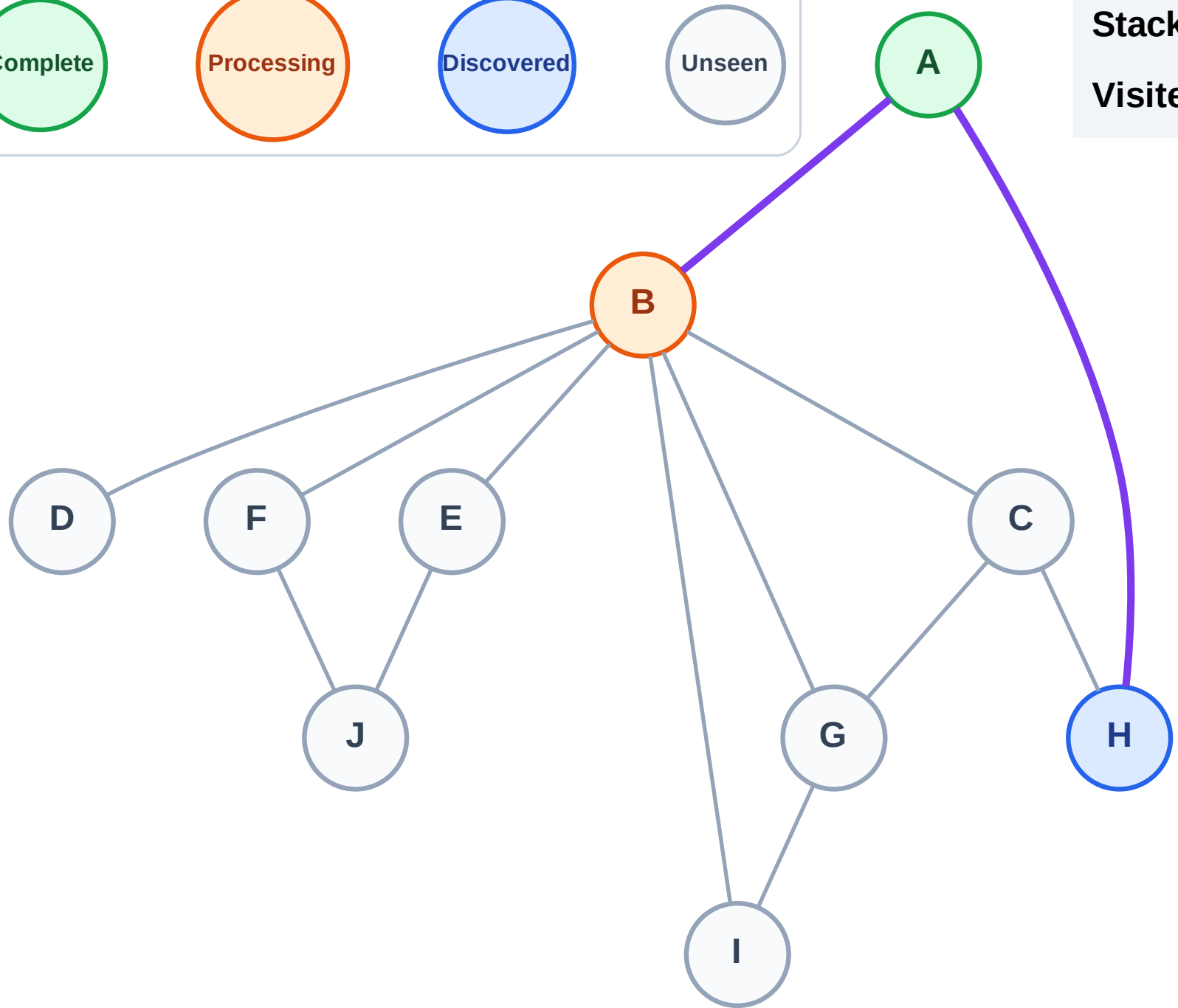
Step 4: Process node B

Legend



Stack (top at right): H

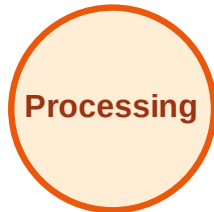
Visited: A



DFS Traversal

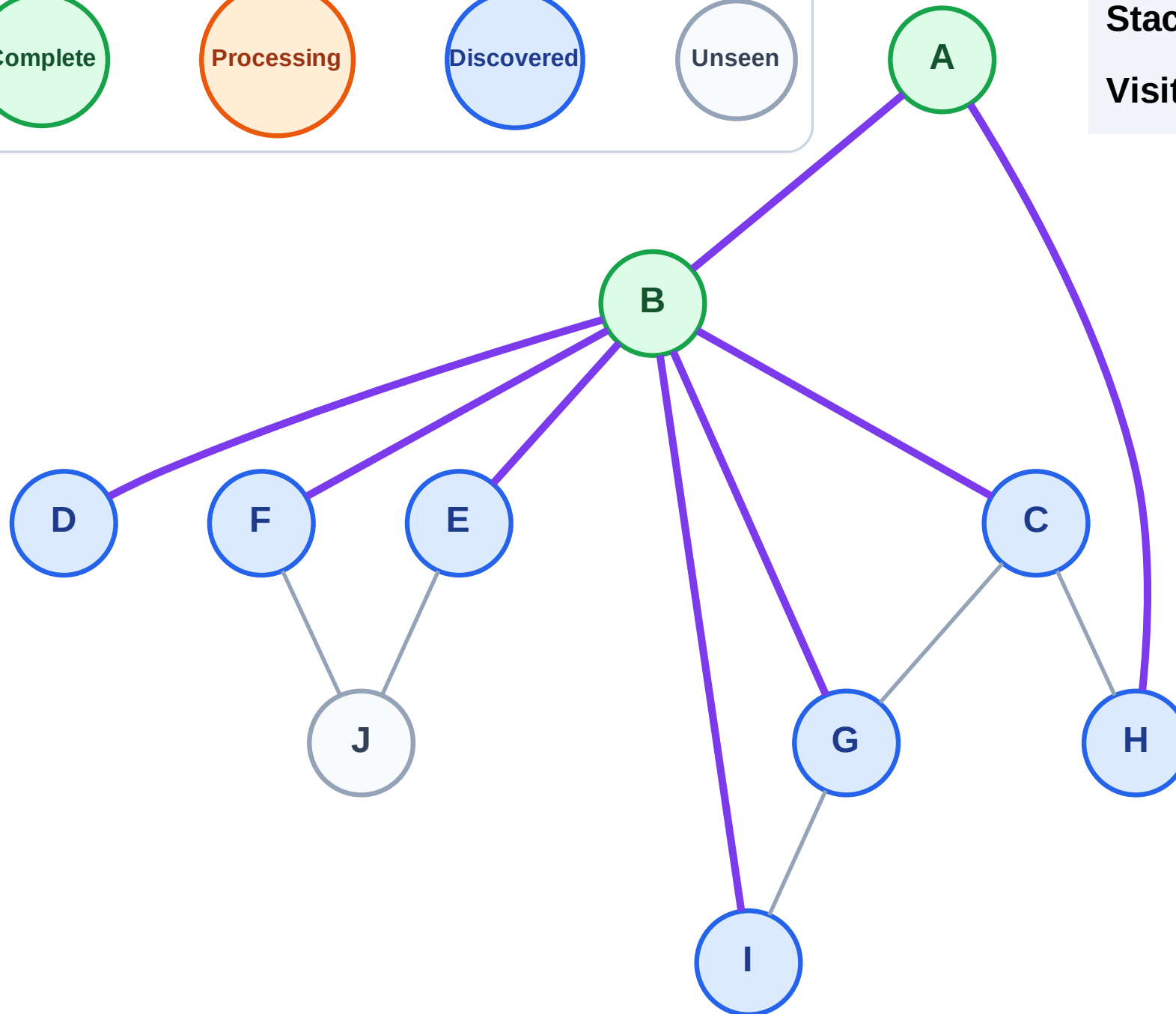
Step 5: Discover I, G, F, E, D, C from B

Legend



Stack (top at right): H | I | G | F | E | D | C

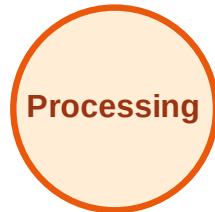
Visited: A, B



DFS Traversal

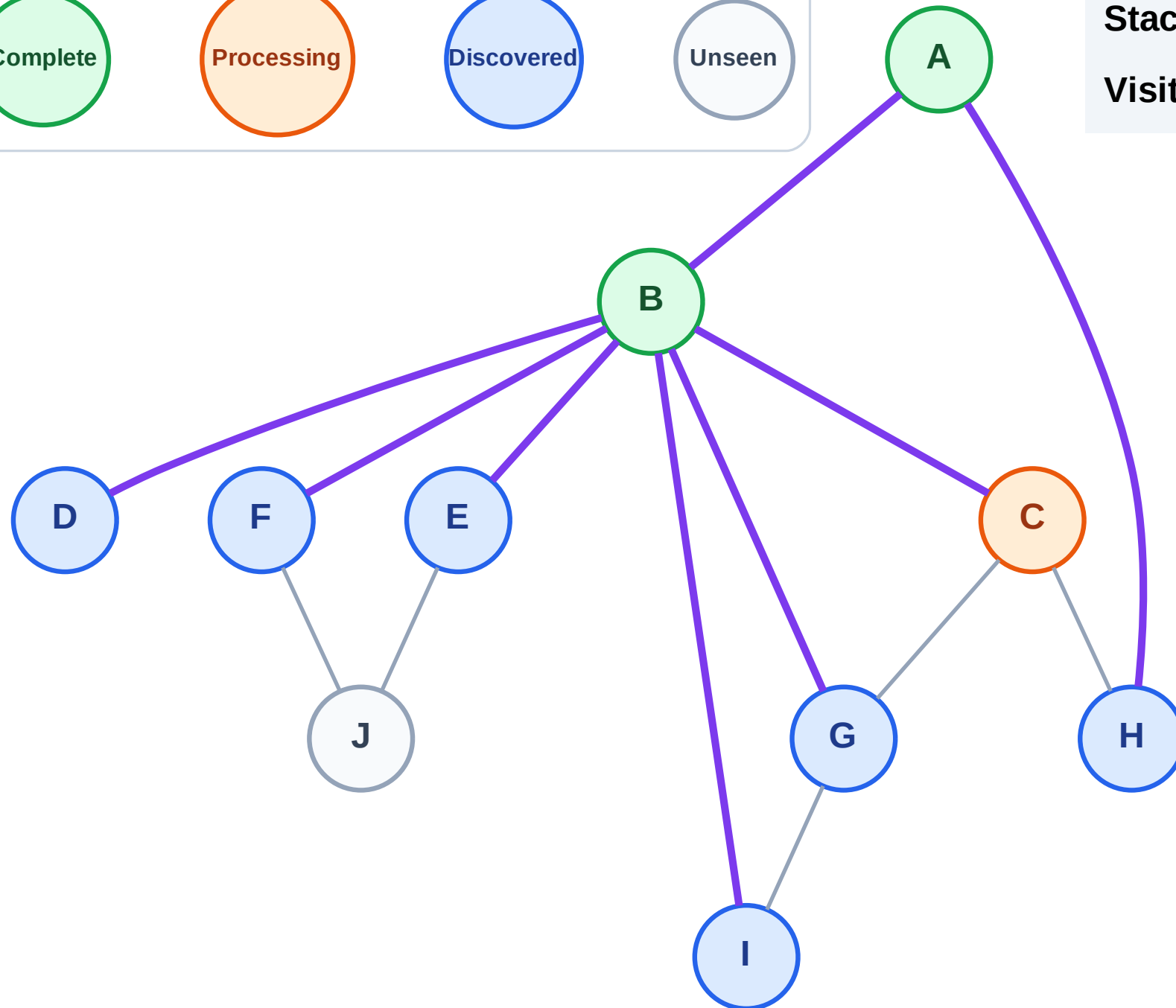
Step 6: Process node C

Legend



Stack (top at right): H | I | G | F | E | D

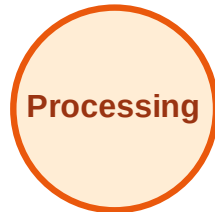
Visited: A, B



DFS Traversal

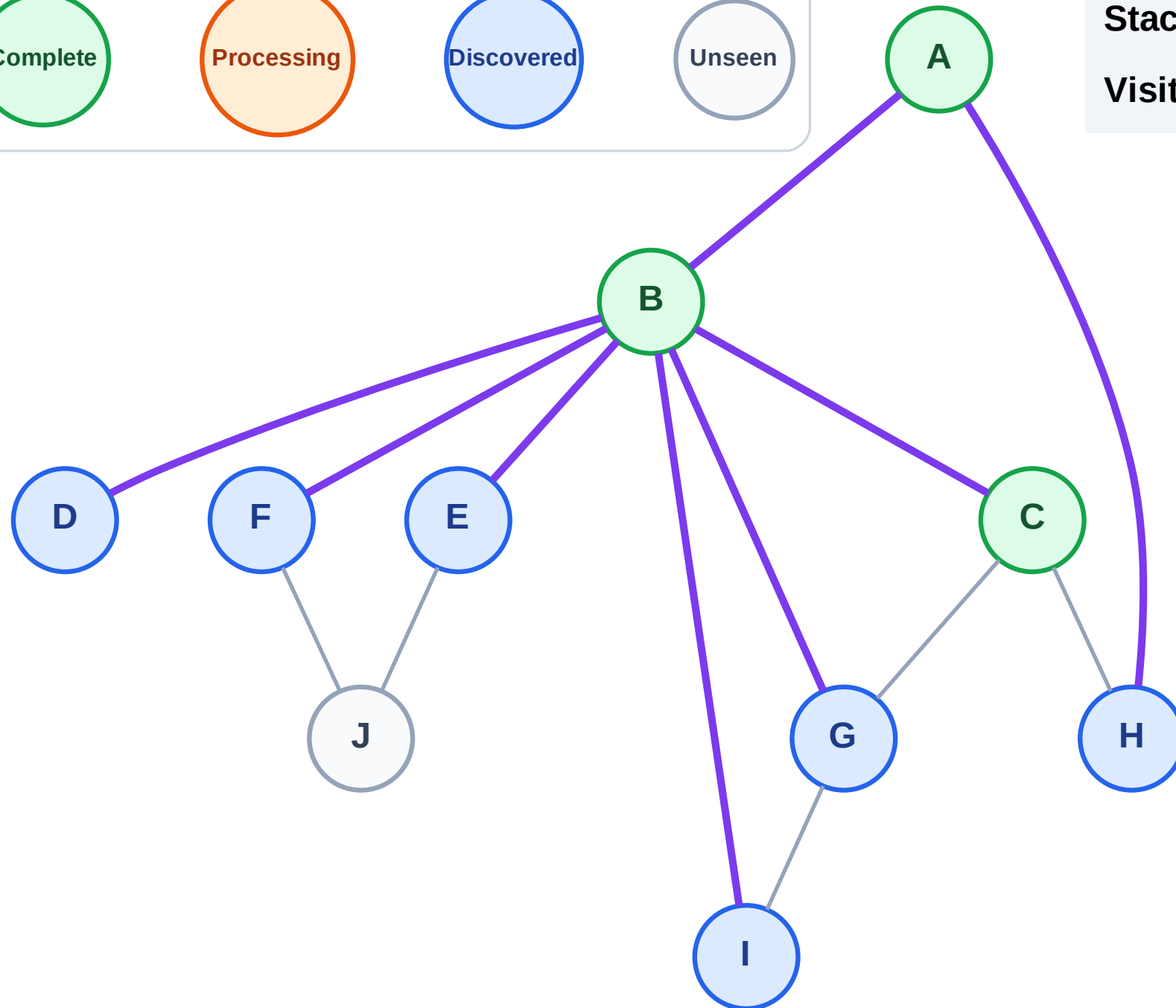
Step 7: No new neighbors from C

Legend



Stack (top at right): H | I | G | F | E | D

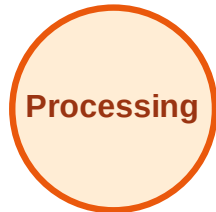
Visited: A, B, C



DFS Traversal

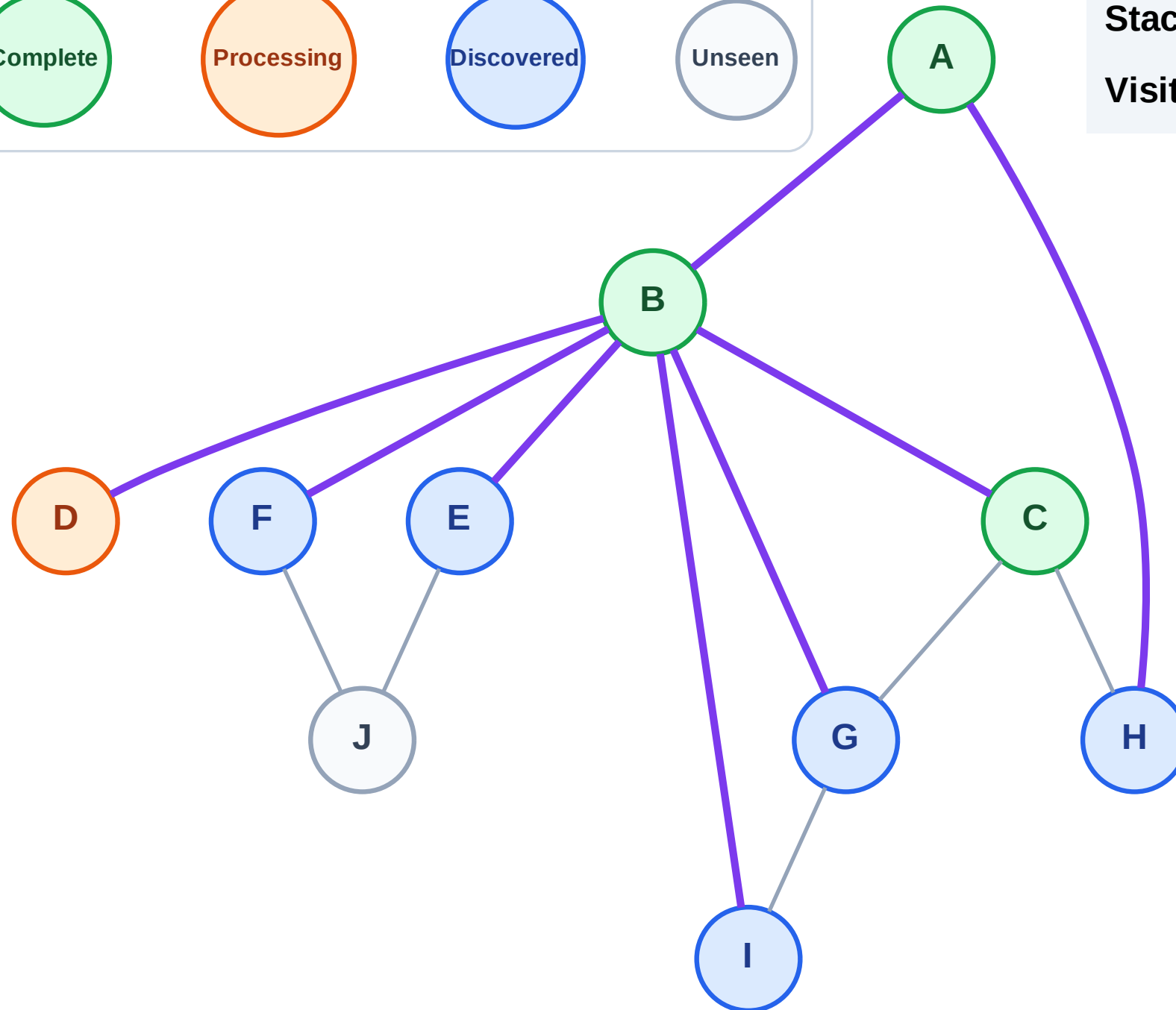
Step 8: Process node D

Legend



Stack (top at right): H | I | G | F | E

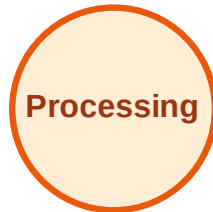
Visited: A, B, C



DFS Traversal

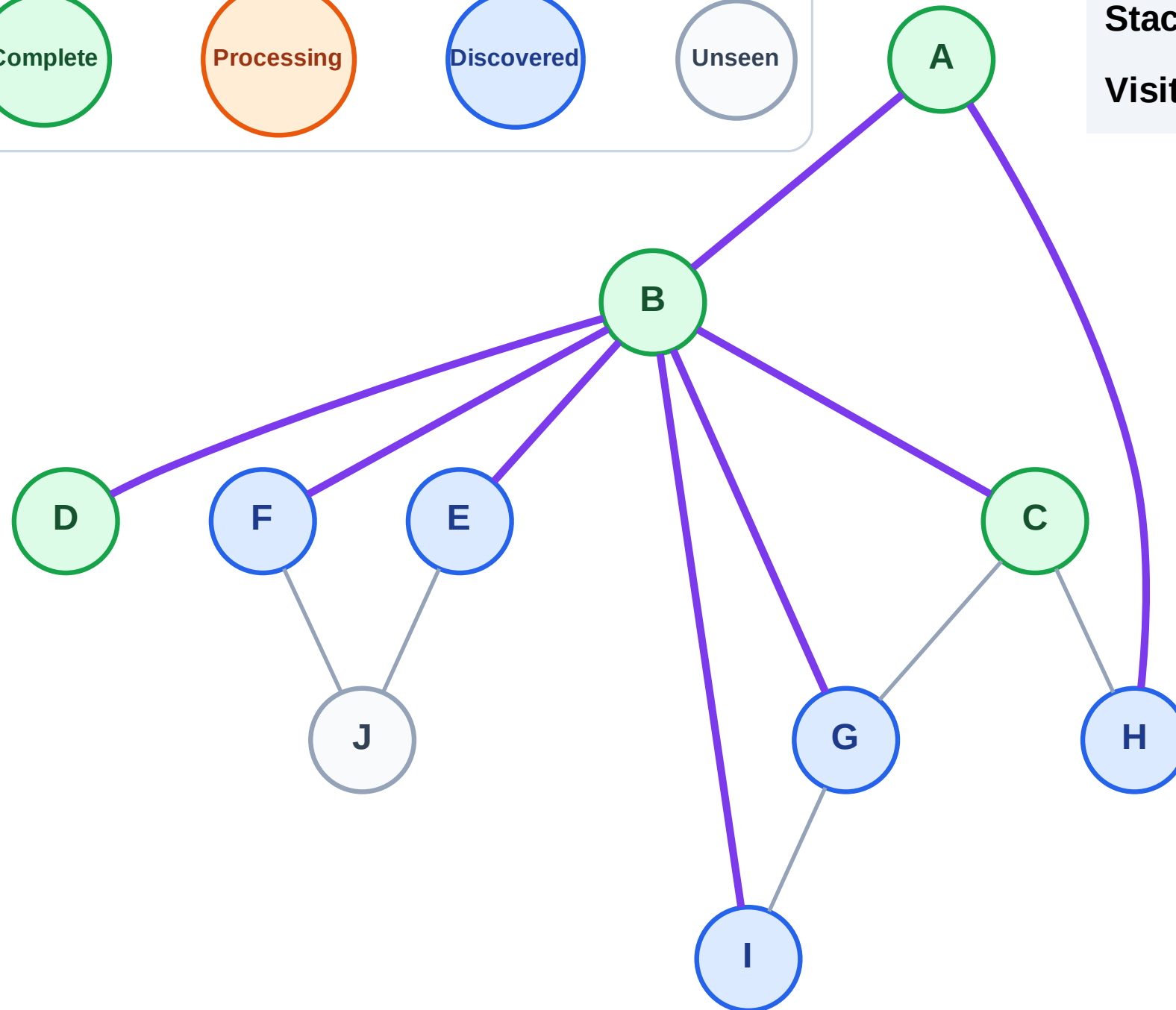
Step 9: No new neighbors from D

Legend



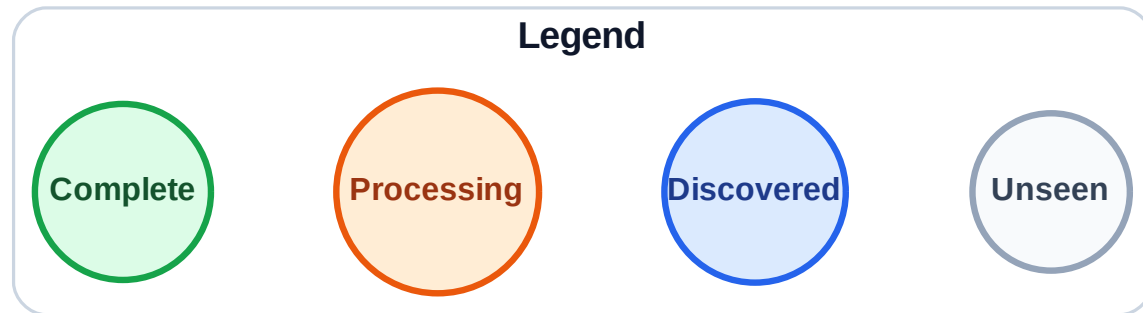
Stack (top at right): H | I | G | F | E

Visited: A, B, C, D



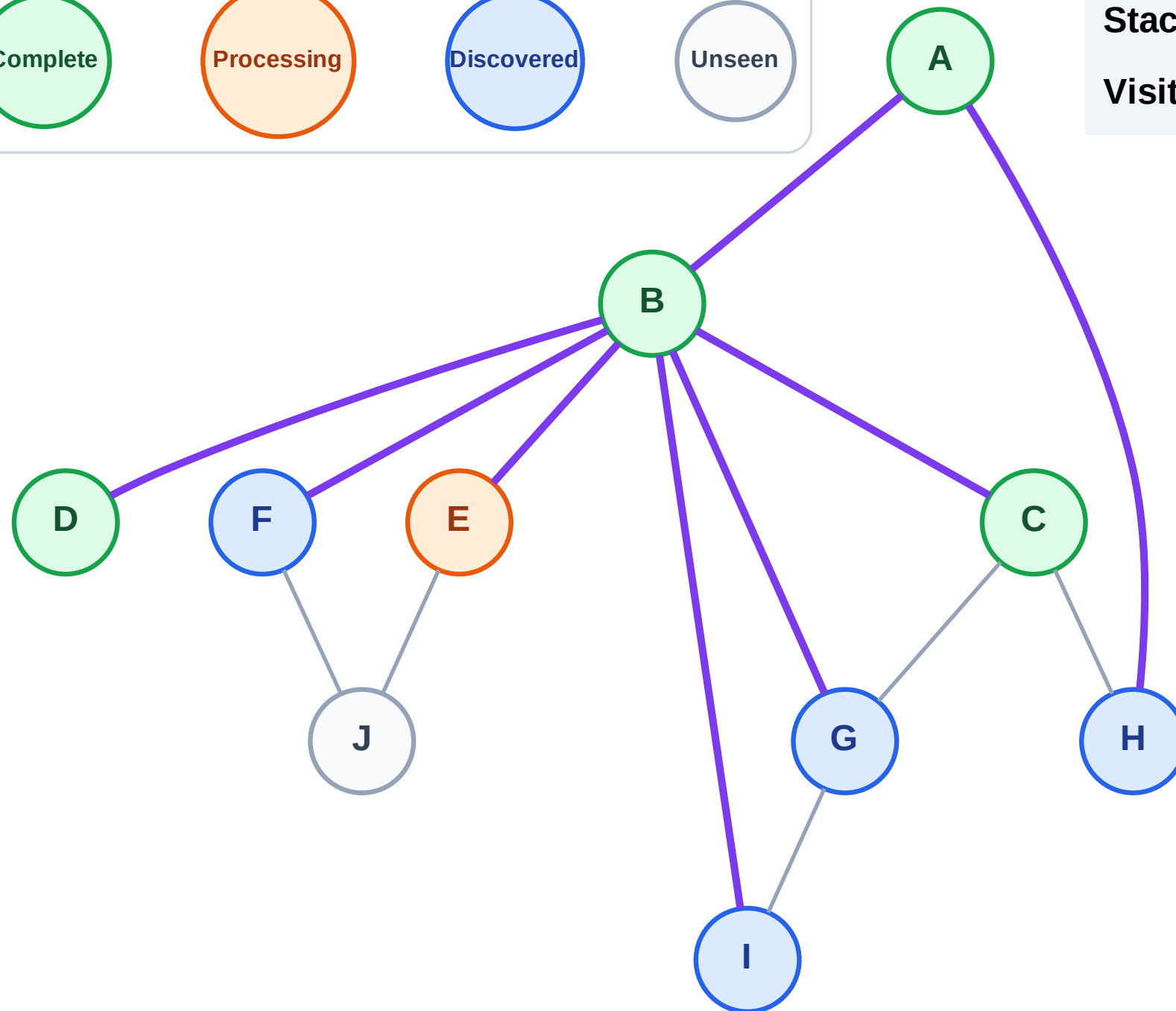
Step 10: Process node E

Step 10: Process node E



Stack (top at right): H | I | G | F

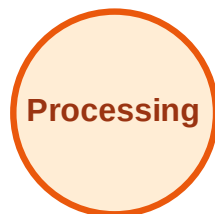
Visited: A, B, C, D



DFS Traversal

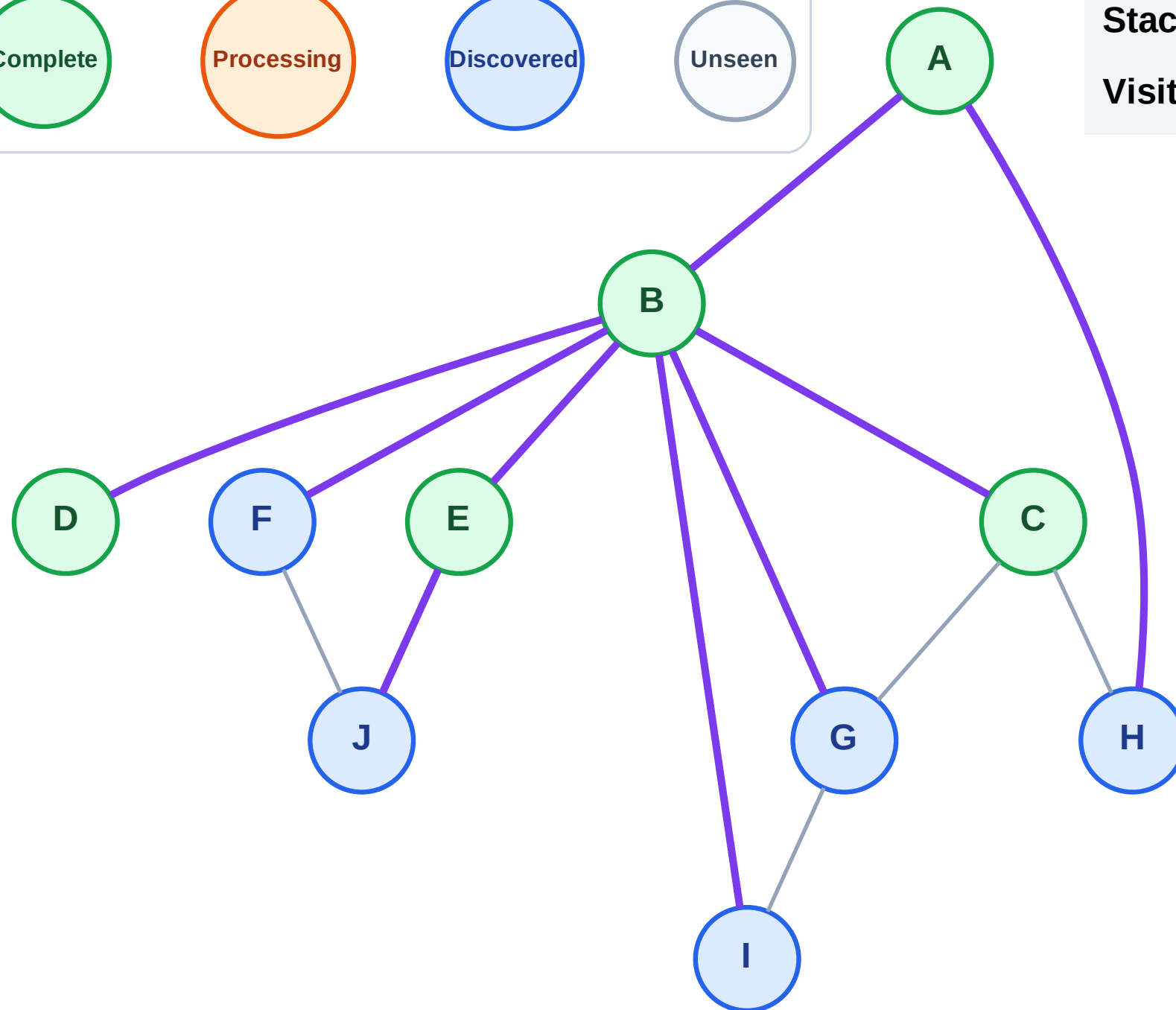
Step 11: Discover J from E

Legend



Stack (top at right): H | I | G | F | J

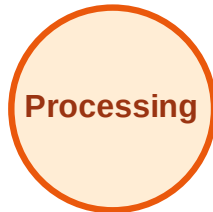
Visited: A, B, C, D, E



DFS Traversal

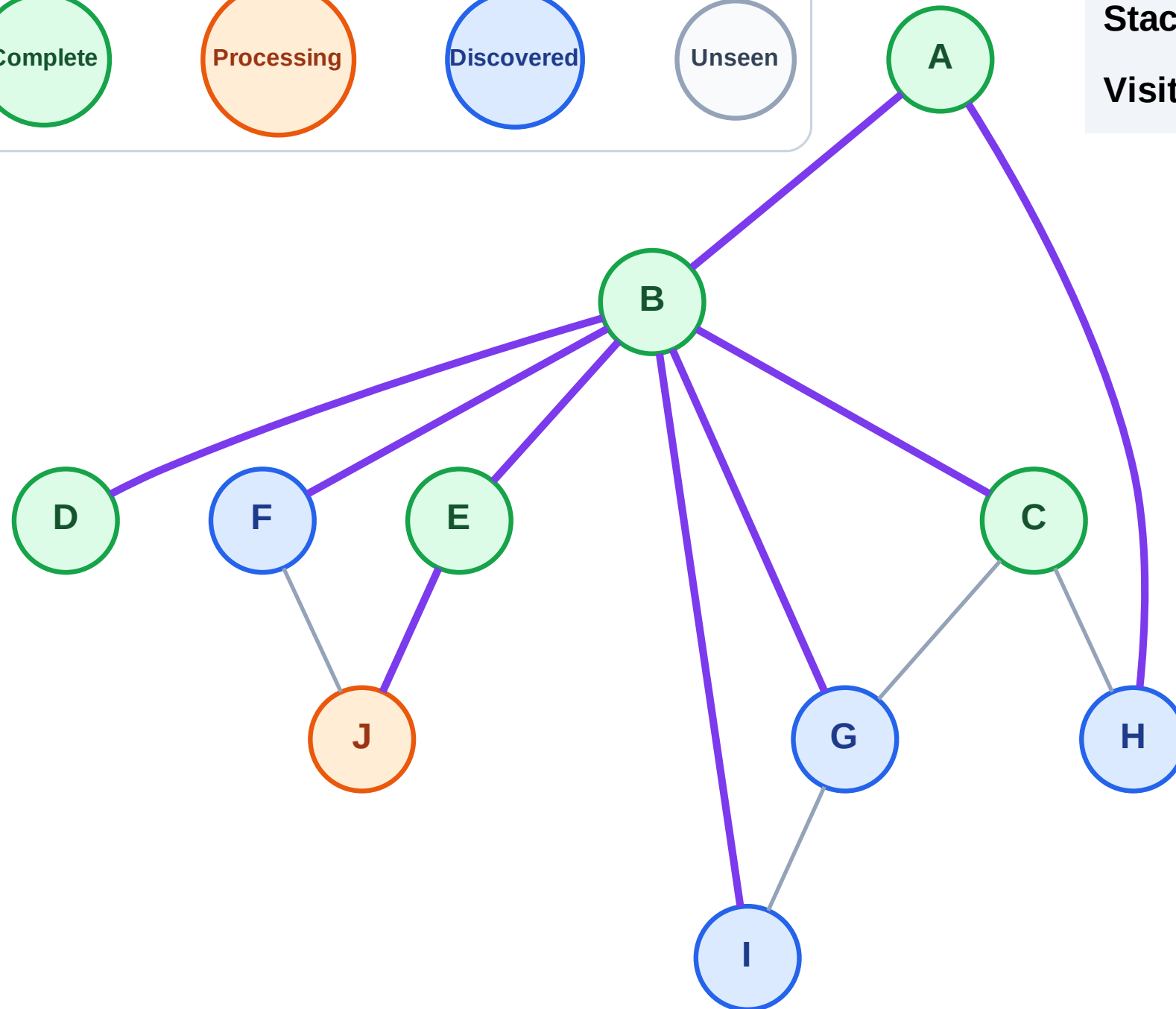
Step 12: Process node J

Legend



Stack (top at right): H | I | G | F

Visited: A, B, C, D, E



DFS Traversal

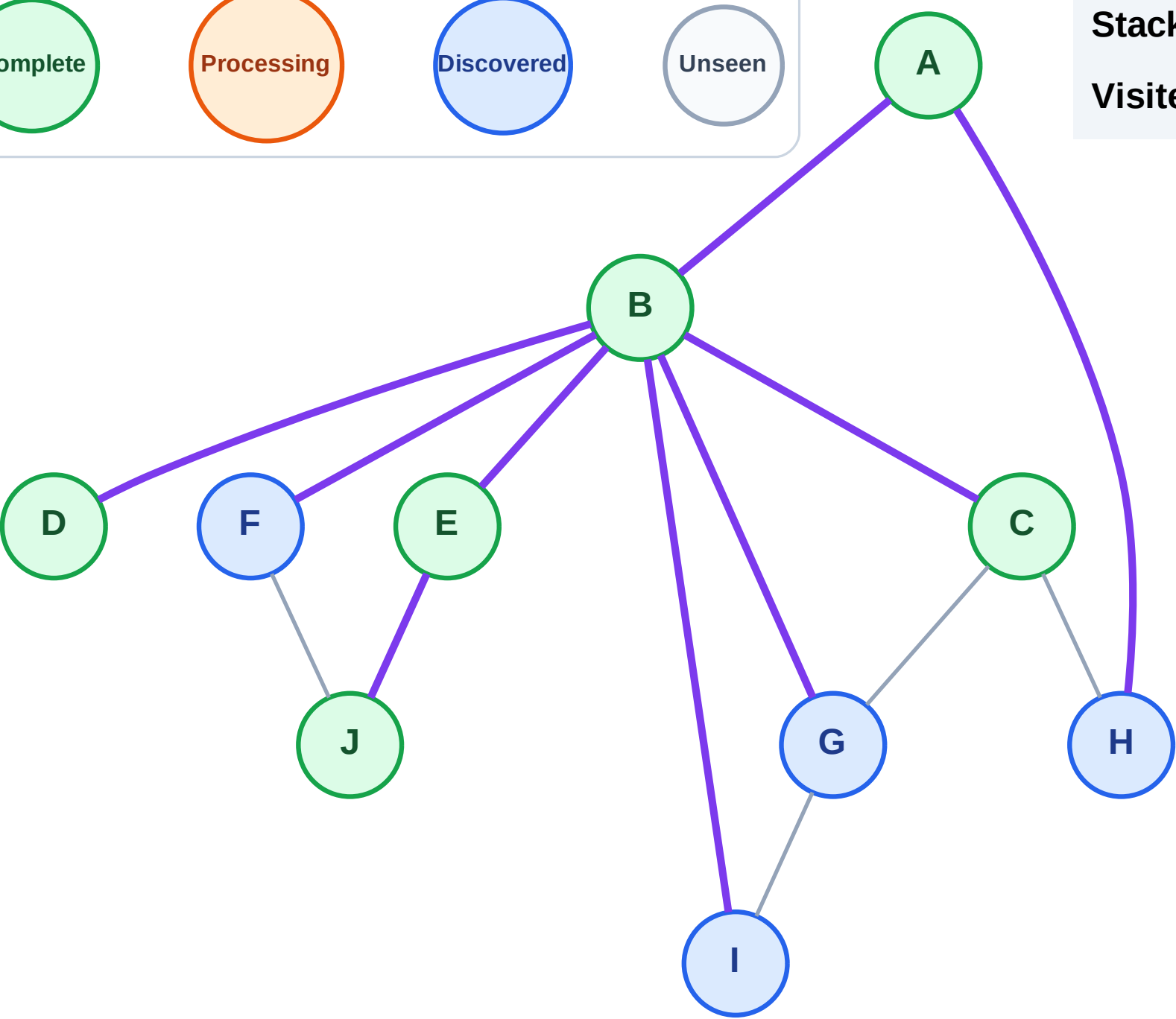
Step 13: No new neighbors from J

Legend



Stack (top at right): H | I | G | F

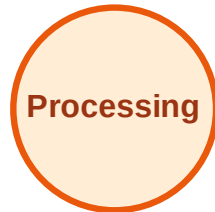
Visited: A, B, C, D, E, J



DFS Traversal

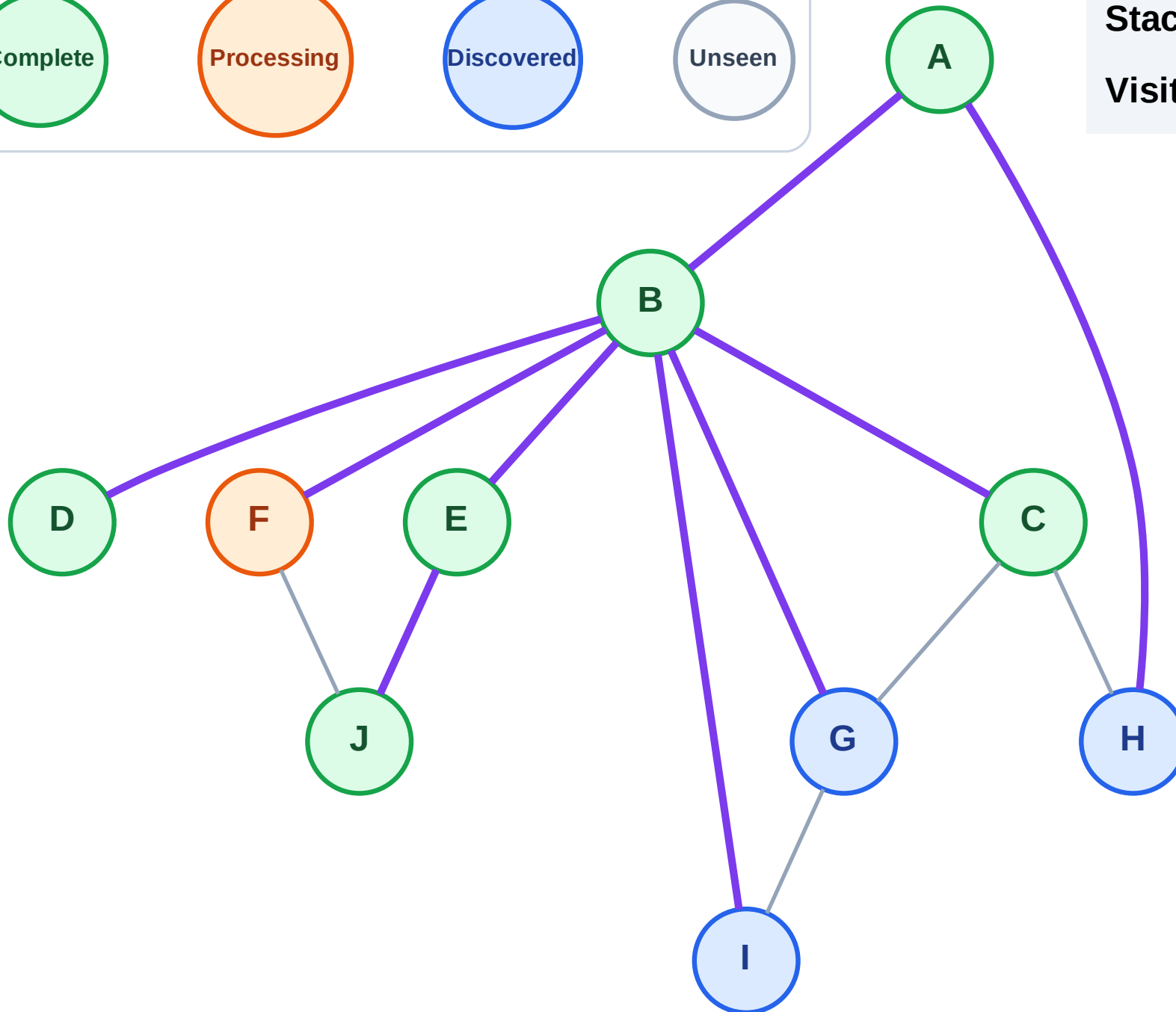
Step 14: Process node F

Legend



Stack (top at right): H | I | G

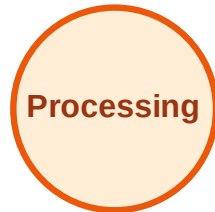
Visited: A, B, C, D, E, J



DFS Traversal

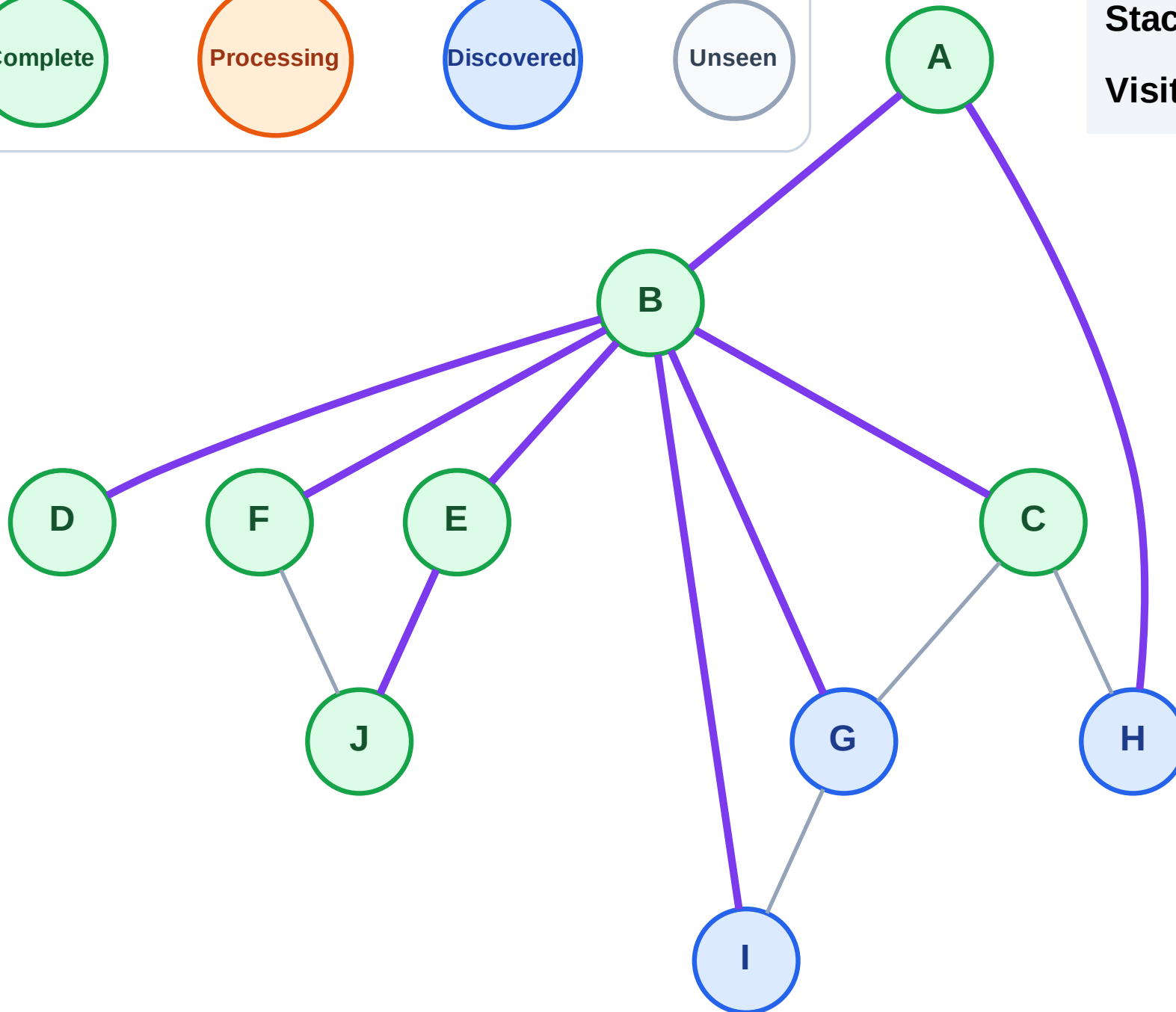
Step 15: No new neighbors from F

Legend



Stack (top at right): H | I | G

Visited: A, B, C, D, E, F, J



DFS Traversal

Step 16: Process node G

Legend

Complete

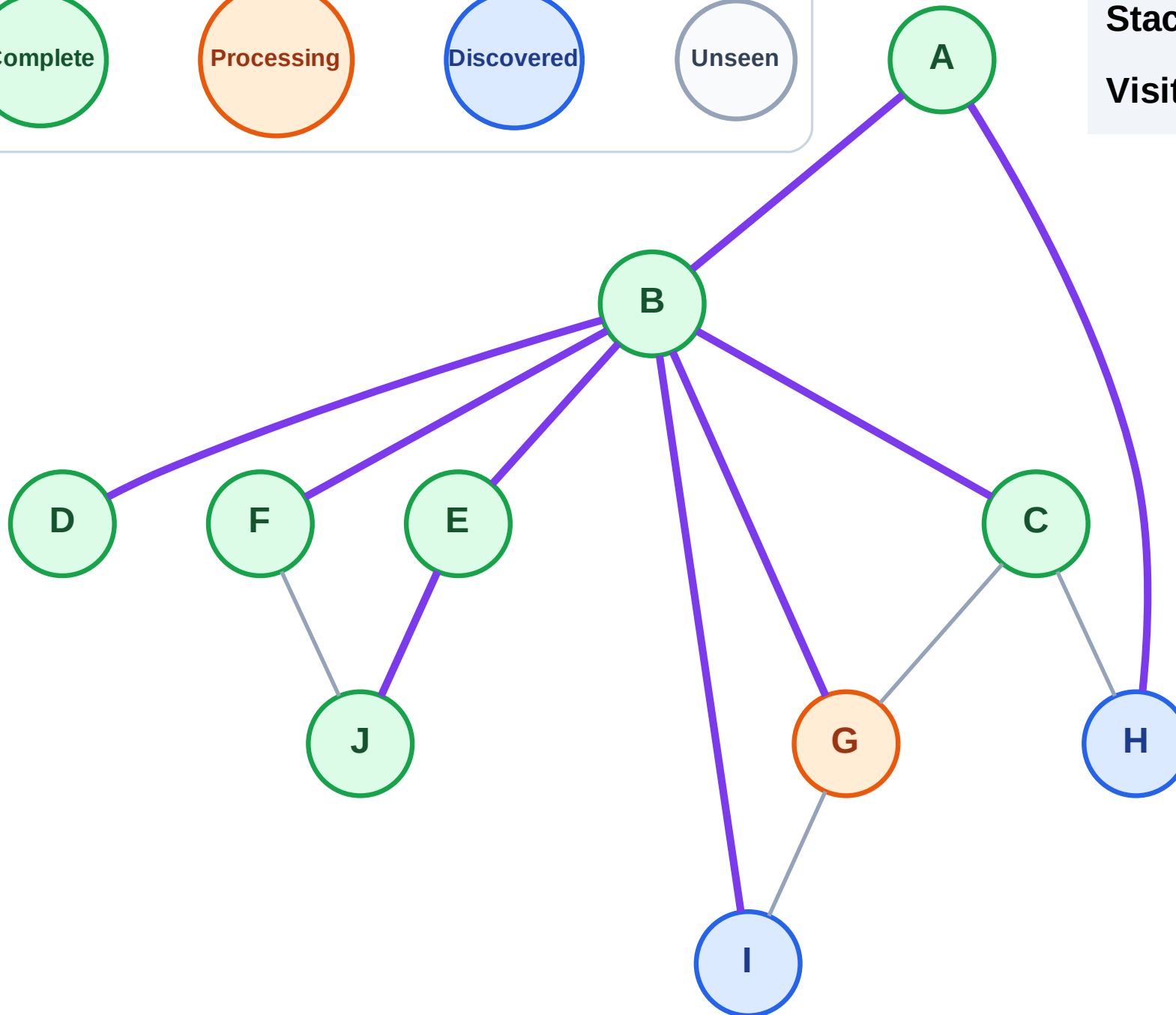
Processing

Discovered

Unseen

Stack (top at right): H | I

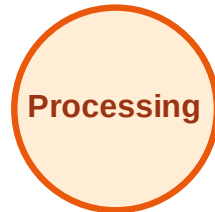
Visited: A, B, C, D, E, F, J



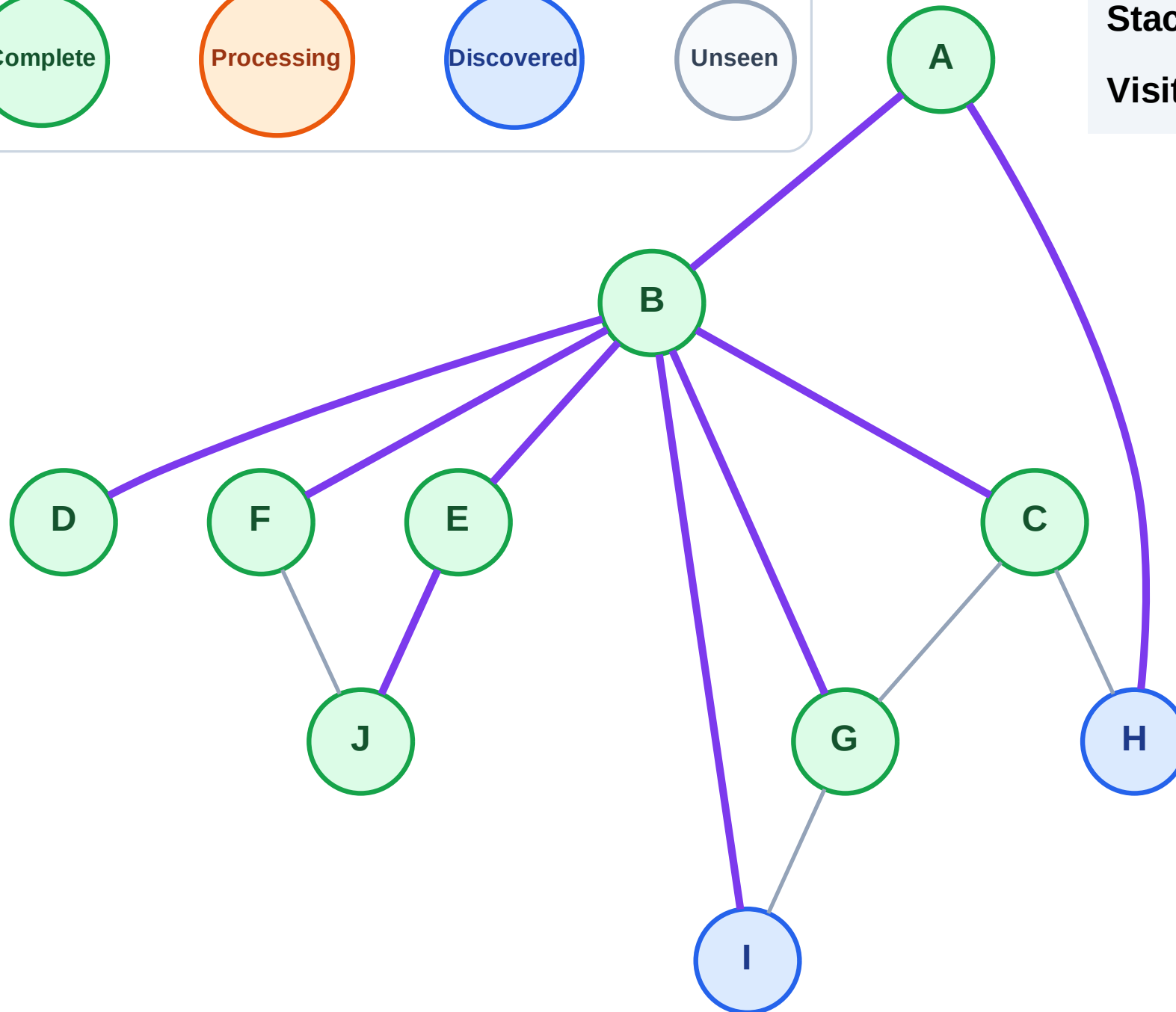
DFS Traversal

Step 17: No new neighbors from G

Legend



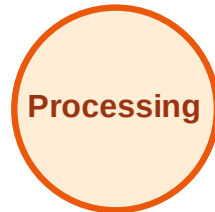
Stack (top at right): H | I
Visited: A, B, C, D, E, F, G, J



DFS Traversal

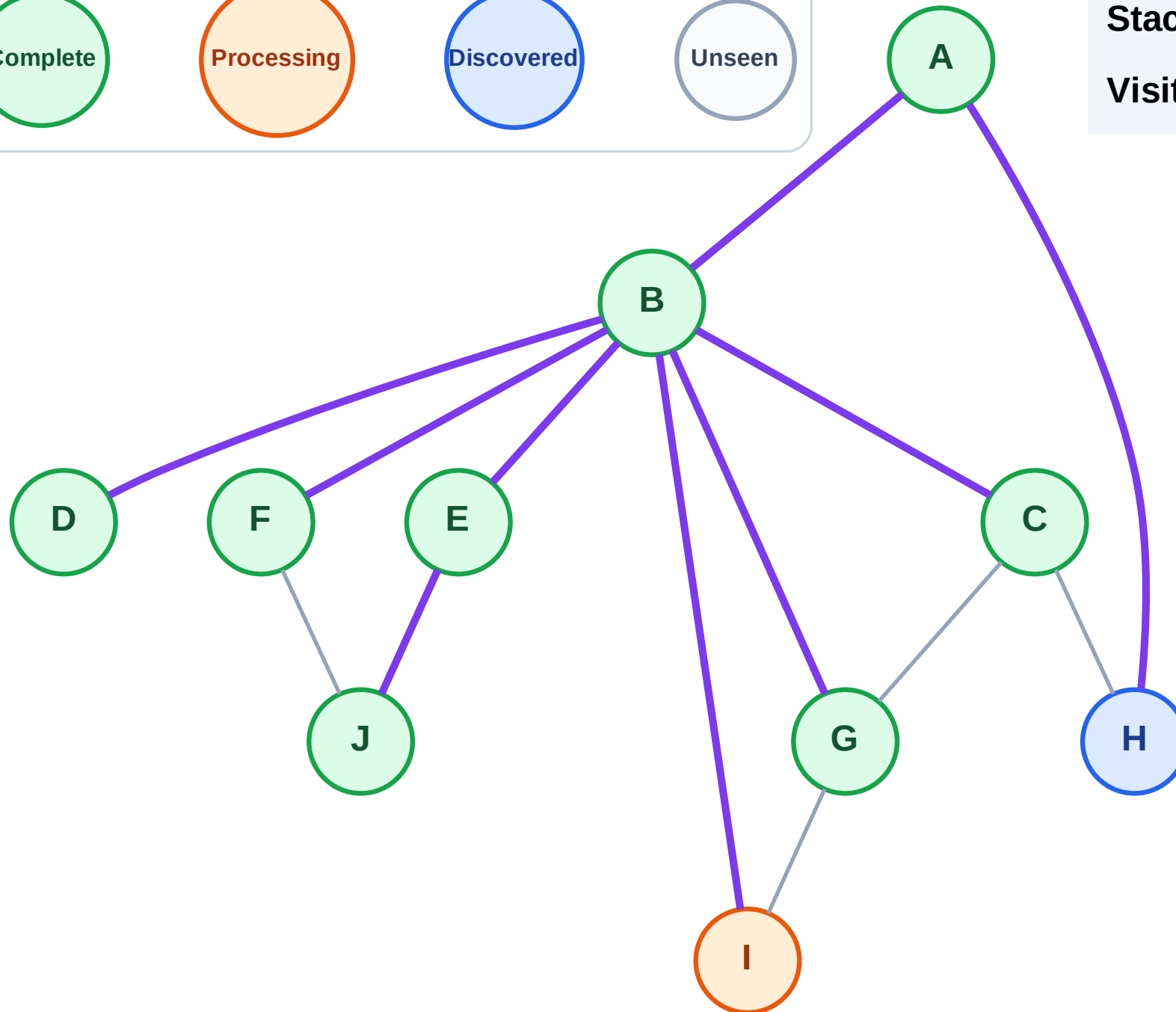
Step 18: Process node I

Legend



Stack (top at right): H

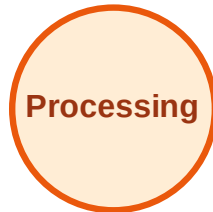
Visited: A, B, C, D, E, F, G, J



DFS Traversal

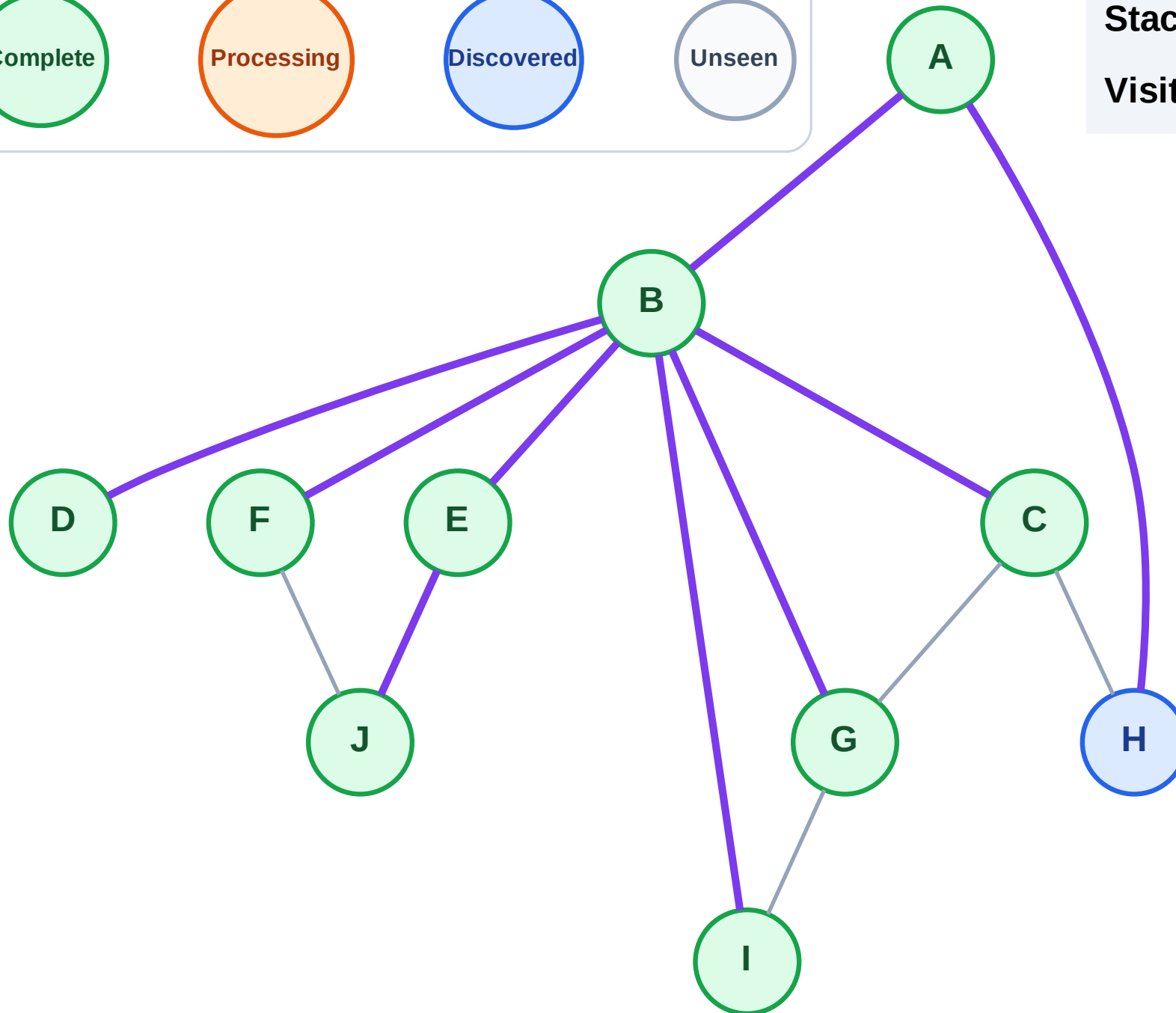
Step 19: No new neighbors from I

Legend



Stack (top at right): H

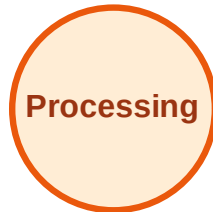
Visited: A, B, C, D, E, F, G, I, J



DFS Traversal

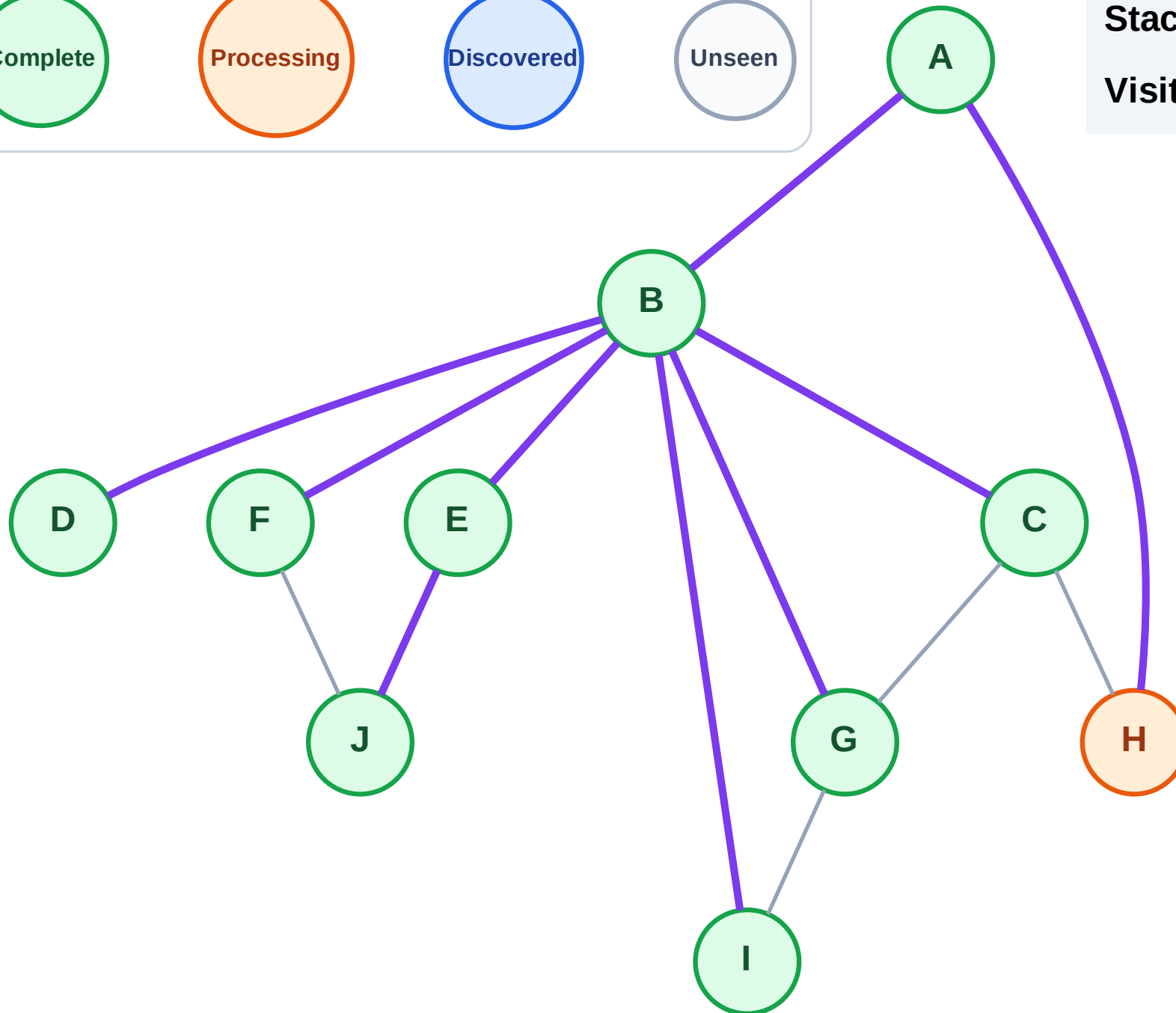
Step 20: Process node H

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, I, J



Step 21: No new neighbors from H

Legend



Visited: A, B, C, D, E, F, G, H, I, J

